

Diversion Risk, Markups and the Financing Cost Advantage of Trade Credit*

Alvaro Garcia-Marin[†] Santiago Justel[‡] Tim Schmidt-Eisenlohr^{§, ¶}

February 2023

Abstract

Trade credit is the most important form of short-term finance in both emerging and developed economies. This paper develops a model where trade credit reduces borrowing from banks. This gives rise to a financing cost advantage of trade credit that increases in the product of markups and borrowing costs. In line with this prediction, Chilean export data show that a one-standard-deviation rise in upstream markups increases trade credit by 13 days. The extensive and intensive margins contribute about equally to this effect, which strengthens with the destination country's borrowing costs. These findings are robust to instrumenting markups with estimated physical productivity.

Keywords: Trade credit, financing costs, financial intermediation

JEL Classification: F12, F14, G21, G32

*We would like to thank JaeBin Ahn, Julia Cajal Grossi, Tore Ellingson, Ryan Kim, Fernando Leibovici, Volodymyr Lugovskyy, Evren Örs, Mitchell Petersen, as well as seminar audiences at the Graduate Institute Geneva, ETH Zurich, University of Bayreuth, George Washington University, Universidad de Chile, Universidad Alberto Hurtado, Universidad de Santiago, University of Mainz, and the IDB Research Seminar, and conference participants at the CESifo Conference on Global Economy, EIIT Conference, FIRS Annual Conference, FDIC Bank Research Conference, LACEA-LAMES, Mid-Atlantic International Trade Workshop, Midwest Trade Conference, SED Conference, and RIDGE Conference, for helpful comments and suggestions. The views expressed are the authors' and do not necessarily represent the views of the Federal Open Market Committee, its principals, or the Board of Governors of the Federal Reserve System. Alvaro García-Marin gratefully acknowledges research support from ANID-Chile through project Fondecyt Regular 1191536.

[†]Universidad de los Andes, Chile. Email: agarciam@uandes.cl.

[‡]World Bank. Email: sjustel@ucla.edu.

[§]Federal Reserve Board of Governors. Email: t.schmidteisenlohr@gmail.com.

[¶]Corresponding author. Address: Constitution Ave NW &, 20th St NW, Washington, DC 20551.

1 Introduction

Trade credit is the most important form of short-term finance for firms in emerging and developed countries, including the United States.¹ In the first quarter of 2021, non-financial U.S. firms had about \$5.2 trillion in trade credit outstanding, equaling 24 percent of GDP. Trade credit affects key macroeconomic outcomes like the transmission of monetary policy (Nilsen, 2002; Adelino et al., 2020), economic growth (Demirguc-Kunt and Maksimovic, 2001; Fisman and Love, 2003), the structure of supply chains (Kalemli-Ozcan et al. (2014), Antràs (2023), Kim and Shin (2023)), macroeconomic stability (Hardy et al. (2022)), and corporate default (Jacobson and von Schedvin, 2015; Barrot, 2016; Amberg et al., 2021).

Two central questions in the literature are why trade credit coexists with bank lending and what factors make trade credit preferable over advance- or on-time payment. This paper speaks to these questions by showing that trade credit is desirable because it requires less bank credit than cash in advance and therefore generates less diversion risk and lower financing costs. Importantly, in contrast to the previous literature, this financing motive for trade credit does not require differences in financing costs (e.g. Schwartz, 1974) or excess liquidity (Emery, 1984) between buyers and sellers, differences in information about buyers (e.g. Biais and Gollier, 1997) or the ability to recover value from collateral (Mian and Smith, 1992) between sellers and banks, or differences in firms’ ability to divert goods and cash (Burkart and Ellingsen, 2004). While the financing motive proposed here does not require these other frictions, adding them back in would further increase the advantage of trade credit. We therefore see the proposed mechanism as complementary to these earlier explanations.

In the model, an upstream firm (the seller) produces and sells a good to a downstream firm (the buyer) at a markup, settling the transaction either through cash in advance or trade credit.² Under cash in advance, the buyer needs to pre-pay the full amount to the seller, which requires borrowing an amount equal to the full invoice. In contrast, extending trade credit requires less bank credit, as the seller only needs to cover production costs in advance which are lower than the sales price if there is a markup. Diversion risk generates a spread between the

¹Trade credit is defined as the implicit lending by a seller to a buyer when a buyer is given some time to pay for goods after receiving them.

²In model extensions, we also allow for partial pre-payments and letters of credit, a payment form used in international trade transactions. The letter of credit extension is particularly relevant as we test the model using international trade data.

borrowing rate and the deposit rate. Therefore, the difference in borrowing needs between cash in advance and trade credit affects profits. The larger the markup and the difference between the borrowing and the deposit rate, the more attractive is trade credit and the longer is the optimal trade credit maturity. All else equal, trade credit is preferred over cash in advance if there is a positive markup and a positive interest rate spread.

Interestingly, the financing motive for trade credit developed here is present even if the seller's borrowing rate exceeds that of the buyer, as long as the seller's deposit rate is below the buyer's borrowing rate. The model is therefore able to explain why it can be optimal from a financial point of view for financially constrained suppliers to extend trade credit to larger and less constrained borrowers, providing a rationale for this puzzling pattern (Murfin and Njoroge, 2015).

While the predictions of the model apply, in principle, to both domestic and international trade transactions, we test them using detailed data on Chilean exports. Specifically, for our analysis, we link two panel data sets. First, Chilean export data, which contain detailed information on the payment choice at the transaction level. Second, Chilean manufacturing survey data that we use to estimate firm-product markups following the method developed by De Loecker et al. (2016).

The data have two advantages for testing the financing cost advantage of trade credit. They include detailed information on all inputs used and output produced by firms, which allows estimating quantity-based markups at the firm-product level and therefore avoids identification problems that arise when markups are estimated with revenue data (De Loecker et al., 2016; Syverson, 2019). And, as payment terms are available for buyers located in many different countries, the analysis can exploit differences in average borrowing costs at the country level, which are independent of firms' payment choices.

To strengthen identification, the empirical analysis does not only look at the direct link between markups and trade credit but also at the effect of the interaction between markups and destination country borrowing costs on trade credit. Studying this interaction term is crucial for two reasons. First, the prediction on markups and borrowing costs is, to our knowledge, unique to the financing cost mechanism presented here. Second, the interaction term allows for the inclusion of a rich set of fixed effects, which directly address omitted variable concerns and make it more plausible that the exclusion restriction for the instrumental variable holds.

In line with the model's predictions, we find that both trade credit use (extensive margin)

and trade credit maturity (intensive margin) increase with markups. Importantly, as predicted by the financing motive, the effect of the markup on trade credit increases with the buyer's borrowing rate. The estimated effects are economically meaningful. A one standard deviation rise in upstream markups increases trade credit, on average, by 13 days, with the extensive and intensive margins contributing about equally to this effect.

Results are robust to alternative measures of markups and the inclusions of a large set of fixed effect and control variables. Results continue to hold when instrumenting markups by plant-product level physical productivity. As the instrument is constructed from technological supply-side factors, the IV estimation also addresses concerns about competition and demand-side effects. U.S. firm-level data from Compustat, which jointly capture both the extensive and intensive margin, are also consistent with the financing cost advantage of trade credit: Trade credit use increases with markups, and this relationship strengthens with higher funding costs, as proxied by the real effective federal funds rate.

Our analysis extends theoretical work on payment choice in international trade ([Schmidt-Eisenlohr, 2013](#); [Ahn, 2014](#); [Antràs and Foley, 2015](#); [Niepmann and Schmidt-Eisenlohr, 2017a](#); [Fischer, 2020](#)), by looking at the interaction between markups and financing costs. It adds to the empirical literature on payment choice in international trade (see e.g. [Antràs and Foley \(2015\)](#), [Demir and Javorcik \(2018\)](#)) by presenting evidence for the proposed financing motive of trade credit. The paper thereby also contributes to the wider literature on financial frictions and trade (see e.g. [Ahn et al., 2011](#); [Amiti and Weinstein, 2011](#); [Manova, 2013](#); [Paravisini et al., 2015](#); [Niepmann and Schmidt-Eisenlohr, 2017b](#); [Leibovici, 2021](#)), which has studied the link between financial conditions, international trade, and the macroeconomy.

The theory also adds to the literature on the dominance of trade credit by providing a new and complementary financing motive of trade credit, namely that trade credit is preferable over cash in advance because it requires less bank credit and hence reduces diversion risk and borrowing costs. Previous papers derived financing motives of trade credit assuming differences in financing costs (e.g. [Schwartz, 1974](#)) or liquidity ([Emery, 1984](#)) between buyers and sellers, or differences in information about buyers (see, e.g. [Biais and Gollier, 1997](#); [Fabbri and Menichini, 2010](#)), or in the ability to recover value from collateral ([Mian and Smith, 1992](#)) between sellers and banks. In contrast, the financing cost channel in our paper is more general and works even if financing costs and liquidity positions are perfectly symmetric, and does not require firms to have any advantage over banks in terms of information about the buyer or in the ability to

recover collateral value to generate a preference for trade credit.

The analysis complements earlier papers with non-financial rationales for the dominance of trade credit. [Burkart and Ellingsen \(2004\)](#), develop a mechanism, where sellers extend trade credit because this type of credit is “in-kind” and is thus harder to divert than cash.³ In our setup, in contrast, diversion is symmetric across bank loans, advance payments, and goods, so that trade credit does not have an advantage because goods are more difficult to divert than cash. [Ferris \(1981\)](#) suggests a model where trade credit serves a transaction motive, separating the exchange of goods from the exchange of money. [Schwartz and Whitcomb \(1979\)](#), [Brennan et al. \(1988\)](#), [Mian and Smith \(1992\)](#), and [Giannetti et al. \(2021\)](#) rationalize trade credit use as a way to price discriminate. [Daripa and Nilsen \(2011\)](#) show that when there is demand uncertainty, an upstream firm may use trade credit to subsidize inventory holdings of a downstream firm. [Chod et al. \(2019\)](#) study an externality when trade credit allows downstream firms to increase their cash purchases from the upstream firm’s competitors. [Wilner \(2000\)](#), [Cuñat \(2007\)](#), [Yang and Birge \(2018\)](#) and [Hardy et al. \(2022\)](#) study the role of trade credit for risk-sharing within a supply chain. None of these theories give rise to the financing cost advantage of trade credit developed here and we are not aware of any model where trade credit use increases in the interaction between markups and the buyer’s borrowing costs.

By showing in micro-data that trade credit use increases with markups, the paper expands on earlier empirical work by [Petersen and Rajan \(1997\)](#), who document a positive correlation between gross profit margins and accounts receivable in a survey data set of small firms. Instead of using gross profit margins at the firm level, this paper estimates markups at the firm-product level, applying the methodology in [De Loecker et al. \(2016\)](#). Furthermore, the detailed transaction-level data allow to separately estimate the effect of markups on the intensive and extensive margins of trade credit, which is not possible when using balance sheet data, as accounts receivable reflect the sum of both margins. Most importantly, by showing that the relationship between markups and trade credit increases in the borrowing rate of the destination country, the paper provides direct evidence for the financing cost advantage of trade credit.

The remainder of the paper is organized as follows. Section 2 presents the model of payment choice and derives the main testable predictions. Section 3 discusses the empirical specifications and presents the methodology for deriving firm-product markups. Section 4 describes our

³[Amberg et al. \(2020\)](#) extend this model with a labor-capital choice, showing that trade credit contributes to a capital bias for financially constrained firms.

dataset. Section 5 presents the main empirical results with Chilean data. Section 6 presents results with U.S. data. Finally, section 7 discusses the implications of our study and routes for future research.

2 A Model of Trade Credit and Markups

This section presents our model of payment choice. It first presents a parsimonious baseline model and derives the main results and testable predictions. It then introduces diversion of bank loans, goods, and advance payments by firms and a competitive banking sector where borrowing rates are endogenous. In a next step, it derives results on trade credit maturity – the number of days granted to buyers until payment. Finally, the baseline model is further generalized in several ways, introducing variable markups, partial pre-payments, Nash-Bargaining, and letters of credit.

The model has two key elements. First, to pay for goods or production costs, firms need to borrow funds from the financial sector. Firms can also deposit surplus liquidity as deposits with the banking sector.⁴ Importantly, banks charge a higher interest rate when lending funds to firms than the interest rates they pay to depositors, as borrowers may divert the borrowed funds. Second, sellers charge markups over marginal costs when selling their goods.

2.1 Baseline Model

One upstream firm (the seller) is matched with one downstream firm (the buyer). Both firms are risk-neutral. There are two periods. In period 0, the seller produces the goods and sends them to the buyer. In period 1, the buyer sells the goods to a final consumer. Because of this time gap between production and final sale, firms need to agree on payment terms. Firms have two options. First, buyers can pay in advance (cash in advance) before receiving the goods. Second, buyers can pay after delivery (on trade credit). A seller produces output for a total cost of C and sells it to the buyer. The buyer can then sell the goods to final consumers and generate revenues R . To finance their transactions, the seller (buyer) can borrow from banks

⁴We assume that firms do not have excess liquidity ex-ante. However, a seller can have excess liquidity after getting paid by the buyer.

at an interest rate r_b (r_b^*), and deposit surplus funds at banks for a deposit rate of r_d .⁵ For all endogenous variables (profits, payment) we use the sub-index “B” for the buyer and sub-index “S” for the seller.

To simplify the exposition, we make two working assumptions that we later relax in sections 2.2 and 2.4. First, the seller makes a take-it-or-leave-it offer to the buyer, who can choose to accept or reject the offer. Second, firms charge a constant markup to final consumers given by μ so that $R = \mu C$. Throughout the analysis, we focus on the interesting case where the markup, μ , is sufficiently large such that both trade credit and cash in advance generate positive profits, $R > (1 + r_b)C$ and $R > (1 + r_b^*)C$, which implies $\mu > 1 + r_b$ and $\mu > 1 + r_b^*$. Let Π_j^i denote the profit of the buyer or seller ($j \in \{B, S\}$) under trade credit or cash in advance ($i \in \{TC, CIA\}$).

Trade Credit. Under trade credit, the seller maximizes:

$$\begin{aligned}\Pi_S^{TC} &= P^{TC} - (1 + r_b)C, \\ \text{s.t. } \Pi_B^{TC} &= R - P^{TC} \geq 0,\end{aligned}$$

where P^{TC} is the total payment from the buyer to the seller. Under trade credit, the seller gets paid P^{TC} , while incurring the production costs C . Because production takes place in period 0 while sales only occur in period 1, the seller has to borrow the production costs C from a bank and pay the interest rate r_b . The maximization is subject to the participation constraint of the buyer. Solving for the optimal payment, P^{TC} , that respects the participation constraint implies $P^{TC} = R$, delivering profits of:

$$\Pi_S^{TC} = R - (1 + r_b)C. \tag{1}$$

Cash in Advance. Under cash in advance, the seller maximizes:

$$\begin{aligned}\Pi_S^{CIA} &= (1 + r_d)(P^{CIA} - C), \\ \text{s.t. } \Pi_B^{CIA} &= R - (1 + r_b^*)P^{CIA} \geq 0.\end{aligned}$$

⁵The assumption that the seller’s outside option is the deposit rate could be relaxed, as the mechanism works as long as the seller’s marginal return to capital is below the buyer’s borrowing rate.

In period 0 the seller gets paid P^{CIA} and incurs production costs C . If the price charged to the buyer exceeds production costs, the seller deposits the surplus funds at a bank for interest rate r_d . The buyer pays P^{CIA} in period 0, borrowing from a bank at interest rate r_b^* . Solving for the optimal payment, P^{CIA} , that makes the buyer's participation constraint bind delivers $P^{CIA} = \frac{1}{1+r_b^*}R$, with seller profits of:

$$\Pi_S^{CIA} = (1 + r_d) \left(\frac{1}{1 + r_b^*} R - C \right). \quad (2)$$

Optimal Payment Choice. Combining equations (1) and (2) and rewriting shows that trade credit is preferred if:

$$\Pi_S^{TC} - \Pi_S^{CIA} = \left[\mu - (1 + r_b) - (1 + r_d) \left(\frac{1}{1 + r_b^*} \mu - 1 \right) \right] C > 0. \quad (3)$$

For the special case where borrowing rates are equal for both firms ($r_b = r_b^*$), the condition simplifies to:

$$(r_b - r_d)(\mu - (1 + r_b)) > 0. \quad (4)$$

Equation (3) can be rewritten to find that trade credit is preferred as long as:

$$\mu \frac{r_b^* - r_d}{1 + r_b^*} > r_b - r_d. \quad (5)$$

That is, with a sufficiently large markup, firms always prefer trade credit as long as the buyer's borrowing rate exceeds the seller's deposit rate. These findings are summarized in the following proposition.

Proposition 1 (Payment Choice)

Suppose the buyer's borrowing rate is above the seller's deposit rate, $r_b^ > r_d$, and $\mu > 1 + r_b$. Then:*

- i) If the buyer and seller face equal borrowing costs ($r_b = r_b^*$), the seller always prefers trade credit.*
- ii) There is always a markup, μ , that is large enough to make the seller choose trade credit*

over cash in advance.

Proof. Follows directly from equations (4) and (5). ■

The financing friction combined with a positive markup provides a rationale for the dominance of trade credit in firm-to-firm transactions. Trade credit dominates cash in advance because it requires less bank credit and thereby minimizes financial intermediation costs. Importantly, the preference for trade credit does not depend on any financial cost advantage of the seller over the buyer. In fact, as shown in equation (4), trade credit is preferred when buyers and sellers face the same borrowing and deposit rates.

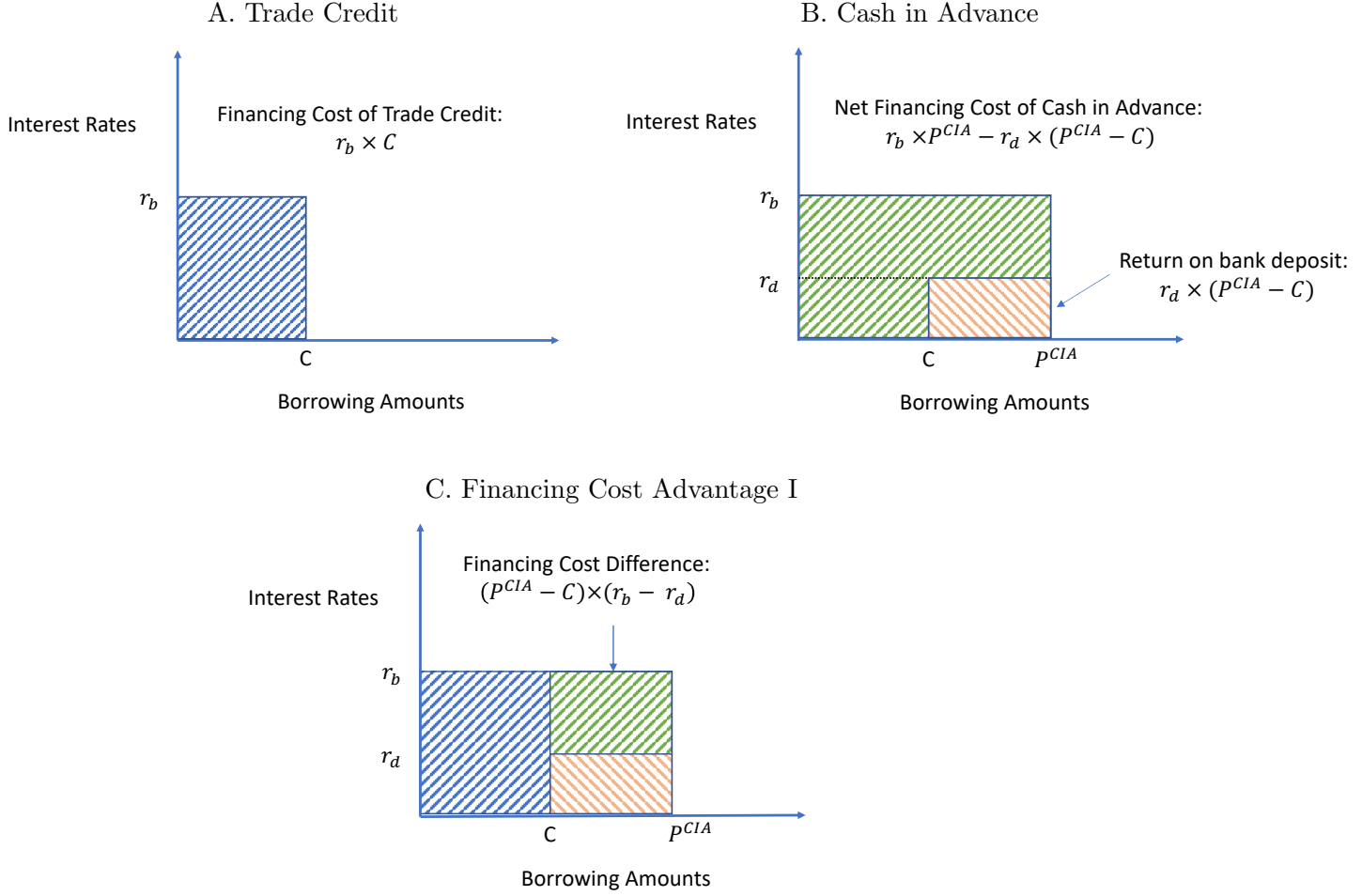
To build further intuition, figure 1 illustrates the financing cost advantage of trade credit for the symmetric case. Panels A and B show the net financing costs for trade credit and cash in advance, respectively, which equal the gross borrowing costs minus any returns on bank deposits. Panel C illustrates the difference in financing costs between cash in advance and trade credit (green area). It is easy to see that $r_b > r_d$ and $\mu > 1 + r_b$ imply a preference for trade credit in the symmetric case, as derived formally in result (i) of proposition 1.

Testable Predictions. To derive testable predictions, take the derivative of equation (3) with respect to μ . Rearranging shows that profits with trade credit relative to cash in advance rise in the markup if:

$$r_b^* - r_d > 0. \tag{6}$$

That is, as long as the buyer's borrowing rate is above the seller's deposit rate, trade credit becomes more attractive relative to cash in advance when the markup goes up. When buyers and sellers face the same borrowing rate, this condition is satisfied as long as financial intermediation is costly and a firm pays more to a bank for borrowing funds than it receives for depositing them. The next section shows that this is indeed the equilibrium outcome when borrowers can divert funds from banks. In the international context, when buyers are located overseas, the condition is more likely to hold if the destination country has a higher borrowing rate and less likely to hold if the source country has a higher deposit rate. Importantly, while in the baseline model we assume that the opportunity cost of cash is given by the deposit rate, our mechanism holds as long as the buyer's borrowing rate exceeds the marginal return to capital of the seller, as this ensures that conditions (5) and (6) are satisfied. The following proposition

Figure 1. The Financing Cost Advantage of Trade Credit



Notes: The figures illustrate the financing cost of trade credit (panel A), cash in advance (panel B), and the difference in financing cost between cash in advance and trade credit (panel C) for the case where the seller and the buyer face the same borrowing and deposit rates. As long as there is a positive spread between borrowing and deposit rate and markups are above $(1 + r_b)$, trade credit has a financing cost advantage over cash in advance.

summarizes our results on trade credit and markups:

Proposition 2 (Trade Credit and Markups)

Suppose $r_b^* > r_d$. Then:

- i) The use of trade credit increases with the markup μ .
- ii) This effect increases with r_b^* and decreases with r_d .

Proof. Follows from equation (6) ■

Part (ii) of proposition 2 represents the key prediction for testing the mechanism proposed in this paper: The financing cost advantage of trade credit should be proportional to the product of the markup and the difference between the buyer’s borrowing rate and the seller’s deposit rate. This prediction is intuitive. The difference in borrowing needs between trade credit and cash in advance only matters if there is a positive difference between the borrowing rate and the deposit rate. Naturally, this effect is larger, the larger is the interest rate difference. Panel C of figure 1 illustrates this finding, as the financing cost advantage, the green area, is the product between the interest rate difference and the markup, as $P^{CIA} - C = (\frac{\mu}{1+r_b} - 1)C$.

2.2 Diversion Risk

We now extend the model and micro-found the spread between the borrowing and deposit rates by introducing the possibility that firms divert funds or goods. Importantly, we allow the same type of diversion to take place between firms that implicitly lend to each other. That is, buyers that receive trade credit can divert goods, and sellers that receive advance payments can divert cash.⁶

Assume that a fraction η (η^*) of sellers (buyers) is reliable; that is, these firms always fulfill their contracts. If a firm is unreliable, it does not fulfill its contract voluntarily but diverts goods or funds whenever it gets the opportunity to do so. Assume that an unreliable firm gets the opportunity to divert goods or funds with probability $1 - \phi$.

Banking Sector There is a competitive banking sector, and banks can borrow at the risk-free interest rate r_d . Banks offer loans to sellers and buyers to finance trade credit and cash in advance transactions, respectively. With probability η , the borrower is reliable, and the loan gets repaid in full, and with probability $1 - \eta$, the borrower is unreliable and diverts a share of $1 - \phi$ of the borrowed funds.⁷ We focus on the case where it is optimal for unreliable firms to imitate reliable firms. In appendix C we show that perfect competition between banks ensures

⁶In contrast to Burkart and Ellingsen (2004), we do not assume that goods are harder to divert than cash. Instead, we assume equal abilities to divert across bank loans, trade credit, and advance payments. While we focus on this symmetric case here, results can also be derived in a model where only bank loans can be diverted or where diversion differs between bank loans and firm-to-firm lending. These results are available upon request.

⁷For tractability, we model diversion in reduced-form similar to Antràs and Foley (2015). However, a generalization of the setting in Burkart and Ellingsen (2004) should yield similar results. In Burkart and Ellingsen (2004) diversion does not happen in equilibrium because banks set optimal incentive-compatible

that the only equilibrium is where banks offer a contract that is accepted by both types. Under trade credit, the seller borrows production cost C from the bank at rate r_b . Zero bank profits imply that the banks funding costs equal the expected repayment amount:

$$(1 + r_d) = (1 + r_b)(\eta + (1 - \eta)\phi) \quad (7)$$

Solving for the borrowing rate delivers:

$$1 + r_b(\tilde{\eta}) = \frac{1 + r_d}{\tilde{\eta}},$$

with $\tilde{\eta} = \eta + (1 - \eta)\phi$. Solving the analog problem for the buyer who borrows $\frac{R}{1+r_b^*}$ under cash in advance delivers:

$$1 + r_b^*(\tilde{\eta}^*) = \frac{1 + r_d}{\tilde{\eta}^*},$$

with $\tilde{\eta}^* = \eta^* + (1 - \eta^*)\phi$. As long as there is some diversion $\eta, \eta^*, \phi < 1$, there is a spread between the borrowing rate and the deposit rate.

The Seller Problem We focus on the case where it is optimal for unreliable firms to imitate reliable firms. In appendix C we show that for a sufficiently high share of reliable firms, η , this pooling case is consistent with optimal behavior by both types of firms. Then, it is sufficient to derive the optimal choice of a reliable firm.

Trade Credit The reliable seller's maximization problem reads:

$$\begin{aligned} \mathbb{E}[\Pi_{RS}^{TC}] &= \tilde{\eta}^* P^{TC} - (1 + r_b(\tilde{\eta}))C \\ \text{s.t. } \mathbb{E}[\Pi_{RB}^{TC}] &= R - P^{TC} \geq 0. \end{aligned}$$

A reliable seller gets paid P^{TC} with probability $\tilde{\eta}^*$, while she incurs production costs C with certainty. The optimal payment does not change compared to the baseline model and remains

contracts to prevent this outcome.

$P^{TC} = R$, delivering expected profits of:

$$E[\Pi_{RS}^{TC}] = \tilde{\eta}^* R - (1 + r_b(\tilde{\eta}))C. \quad (8)$$

Cash in Advance The reliable seller maximizes:

$$\begin{aligned} E[\Pi_{RS}^{CIA}] &= (1 + r_d)(P^{CIA} - C), \\ \text{s.t. } E[\Pi_B^{CIA}] &= \tilde{\eta}R - (1 + r_b^*(\tilde{\eta}^*))P^{CIA} \geq 0. \end{aligned}$$

Under cash in advance, there is now a risk that a buyer is matched with an unreliable seller who may not deliver the goods. Thus, the buyer generates revenues R only with probability $\tilde{\eta}$. Solving for the optimal payment delivers $P^{CIA} = \frac{\tilde{\eta}}{1 + r_b^*(\tilde{\eta}^*)}R$. With expected seller profits of:

$$E[\Pi_{RS}^{CIA}] = (1 + r_d) \left(\frac{\tilde{\eta}}{1 + r_b^*(\tilde{\eta}^*)} R - C \right). \quad (9)$$

This represents the general solution for all sellers, as we assumed that conditions are such that an unreliable seller always imitates a reliable seller (see appendix C for details). Taking the difference between expected profits from trade credit and cash in advance delivers:

$$E[\Pi_{RS}^{TC}] - E[\Pi_{RS}^{CIA}] = \left[\tilde{\eta}^* \mu - (1 + r_b(\tilde{\eta})) - (1 + r_d) \left(\frac{\tilde{\eta}}{1 + r_b^*(\tilde{\eta}^*)} \mu - 1 \right) \right] C > 0. \quad (10)$$

For the special case where enforcement is symmetric ($\tilde{\eta} = \tilde{\eta}^*$), the condition simplifies to:

$$(1 - \tilde{\eta})\tilde{\eta} \left(\mu - \frac{1 + r_b(\tilde{\eta})}{\tilde{\eta}} \right) C > 0, \quad (11)$$

which holds as long as there is some diversion of borrowed funds ($\tilde{\eta} < 1$) and trade is profitable ($\mu - \frac{1 + r_b(\tilde{\eta})}{\tilde{\eta}} > 0$). Taking the derivative of equation (10) with respect to μ and rearranging implies that trade credit gets more attractive in μ if:

$$\tilde{\eta}^*(1 - \tilde{\eta}) > 0. \quad (12)$$

That is, if there is any diversion risk, then trade credit use increases in μ . The intuition is quite straightforward: Diversion creates a wedge between the borrowing and deposit rates,

making gross borrowing from banks costly. As trade credit has lower gross borrowing than cash in advance when there is a positive markup, diversion makes trade credit more attractive. Importantly, the fact that firms can divert goods under trade credit or advance cash payments under cash in advance does not affect this trade-off, as these additional diversion opportunities exactly offset each other. The results from the model with diversion are summarized in the following corollary:

Corollary 1 (Trade Credit, Markups, and Diversion)

Suppose $\mu > 1/\tilde{\eta}$ and there is diversion of cash and goods ($\tilde{\eta} < 1$ and $\tilde{\eta}^ < 1$). Then, all predictions in propositions 1 and 2 continue to hold in the model with diversion.*

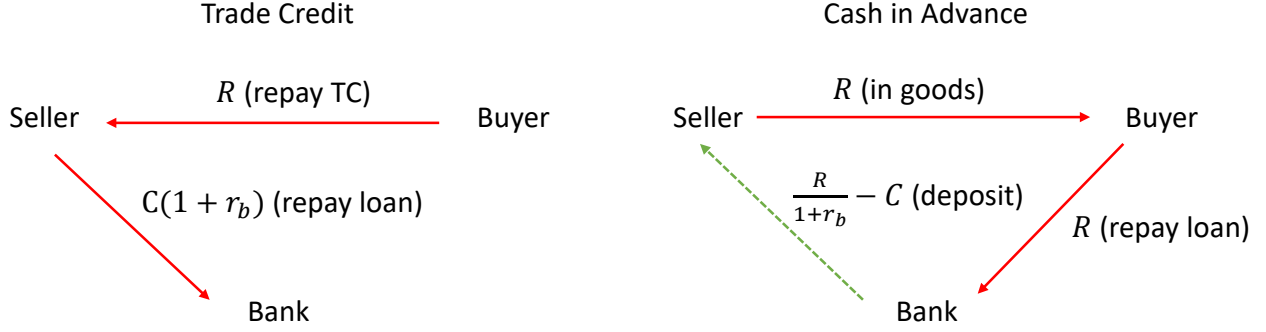
Proof. Proposition 1: i) follows from equation (12); ii) rewrite equation 10 to get $\mu > \frac{(1+r_d)}{\tilde{\eta} \tilde{\eta}^*}$; proposition 2: i) follows directly from equation (12); ii) follows from taking the cross-derivative of equation (10) with respect to μ and r_b^* . ■

We illustrate the intuition for our results on diversion risk in figure 2 and table A.1. The financing cost advantage of trade credit arises because there is a larger total exposure to diversion risk with trade credit than with cash in advance. As shown in the figure and in the first row of the table, firm-to-firm diversion risk is exactly the same for trade credit and cash in advance, with both payment terms implying a value of funds or goods at risk of R . However, the diversion risk arising from bank borrowing is larger with cash in advance, as it requires more gross borrowing (row 2). This leads to higher borrowing costs with cash in advance (row 3) that are only partially offset by the positive deposit returns with cash in advance (row 4) as long as the borrowing rate exceeds the deposit rate (bottom-right cell of the table). Importantly, our results hold even when diverting cash and goods is equally difficult. Making goods harder to divert than cash as in Burkart and Ellingsen (2004) would add an extra factor in favor of trade credit but is not necessary for our mechanism.

2.3 Trade Credit Maturity

When extending trade credit, sellers can decide on the number of days that a buyer has to pay for the goods. We now extend the model to study this intensive margin of trade credit. Let t be the trade credit maturity and let T be the number of days after which revenues are realized by the buyer from selling the goods to final consumers. Assume that diversion risk increases continuously with maturity, t ; a plausible assumption, as longer payment times give rise to

Figure 2. Funds and goods at risk



Notes: This figure shows the funds and goods at risk, represented by the solid red arrows. The left part shows diversion risk under trade credit: the buyer needs to pay R to the seller and the seller needs to pay $C(1 + r_b)$ to the bank. The right part shows diversion risk with cash in advance: the seller has to deliver goods of value R , and the buyer has to pay R to the bank. The figure also shows the deposit of the seller with cash in advance, $\frac{R}{1+r_b} - C$. However, there is no diversion risk for the deposit, so the link is depicted by the green dashed arrow.

more opportunities to divert goods. To capture this, let the probability of payment, $\tilde{\eta}^*(t)$, be a decreasing function of t . The seller then maximizes:

$$\begin{aligned} \mathbb{E}[\Pi_S^{TC,I}] &= (1 + r_d(T - t))(\tilde{\eta}^*(t)P^{TC,I} - C) - r_b t C \\ \text{s.t.: } \mathbb{E}[\Pi_B^{TC,I}] &= R - P^{TC,I}(1 + r_b^*(T - t)) \end{aligned}$$

With an optimal payment of $P^{TC,I} = \frac{R}{1+r_b^*(T-t)}$, this implies expected seller profits of:

$$\mathbb{E}[\Pi_S^{TC,I}] = \left((1 + r_d(T - t)) \left(\frac{\tilde{\eta}^*(t) \mu}{1 + r_b^*(T - t)} - 1 \right) - r_b t \right) C \quad (13)$$

Focusing on the symmetric case and taking the derivative with respect to maturity t , we find that:

$$\begin{aligned} \frac{\partial \mathbb{E}[\Pi_S^{TC,I}]}{\partial t} / C &= \underbrace{(r_b - r_d) \left(\frac{\tilde{\eta}(t) \mu}{(1 + r_b(T - t))^2} - 1 \right)}_{\text{Financing Cost Effect} > 0} \\ &+ \underbrace{\frac{\mu}{1 + r_b(T - t)} [(1 + r_d(T - t))(\tilde{\eta}(t)')]}_{\text{Diversion Effect} < 0} \end{aligned} \quad (14)$$

There are two opposing forces in the model. First, if markups are high enough and the borrowing rate exceeds the deposit rate, then the financing cost advantage of trade credit lets profits rise with maturity, t . Second, as long as the diversion risk increases with maturity, t ,

($\tilde{\eta}(t)' < 0$) the diversion effect lets profits decline with maturity t . For an interior solution to exist, we assume that diversion risk is convex in maturity t and rises sufficiently quickly to dominate the financing cost channel eventually. That is, assume that $\tilde{\eta}(t > 0)'' < 0$ and $-\frac{\tilde{\eta}(t)'}{\tilde{\eta}(t)} > \frac{r_b}{1+r_b(T-t)}$. For the diversion effect to dominate the financing cost effect at T , it is necessary that $\tilde{\eta}(T)' < -\frac{r_b-r_d}{\mu}(\tilde{\eta}\mu - 1)$.⁸ We can then derive the following proposition:

Proposition 3 (Trade Credit Maturity)

Suppose the buyer and the seller face equal borrowing costs and diversion risks ($r_b = r_b^$ and $\tilde{\eta} = \tilde{\eta}^*$), $r_b > r_d$ and $\mu > 1/\tilde{\eta}$. Then,*

- i) there is an optimal level of t that lies between 0 and T*
- ii) the optimal maturity, t , increases in μ , an effect that increases in r_b .*

Proof. See appendix D.1. ■

Proposition 3 implies very similar testable predictions for trade credit maturity as for the payment terms choice. In particular, it predicts that the maturity of trade credit increases in the markup and that the effect of the markup on the maturity increases in the borrowing cost. We illustrate these findings in figure B.1, which plots the optimal maturity, t , against the markup, μ , for different borrowing rates, r_b , setting $\tilde{\eta}(t) = 1 + (\tilde{\eta} - 1) \left(\frac{t}{T}\right)^n$.⁹

2.4 Model Extensions

The following subsections discuss four additional model extensions: Variable markups, partial pre-payments, bargaining between buyers and sellers, and letters of credit.

2.4.1 Variable Markups

Variable markups are a key extension, as they micro-found the instrumental-variable approach employed later, where markups are instrumented with productivity estimates. Assume that firms face a linear demand.¹⁰ Then, the following proposition can be derived, with details in appendix D.2:

⁸To map the model back to our baseline, we assume that there is no diversion risk with zero maturity, $\tilde{\eta}(0) = 1$, and that diversion risk is the same as in the baseline model with maximum maturity, $\tilde{\eta}(T) = \tilde{\eta}$.

⁹In the example, we set $\tilde{\eta} = 0.98$, $r_d = 0.02$, $n = 2$, and $T = 180$.

¹⁰A linear demand would follow, for example, from a demand system as in Melitz and Ottaviano (2008). However, the below results do not depend on this specific modeling choice for variable markups.

Proposition 4 (Trade Credit and Variable Markups)

Suppose the buyer's and seller's borrowing rates are not too different, the seller's borrowing rate is above the seller's deposit rate, $r_b > r_d$, and firms face a linear demand. Then:

- 1. The markup decreases with the marginal cost of production c .*
- 2. By decreasing the markup, an increase in the marginal cost of production makes trade credit less attractive relative to cash in advance.*
- 3. The marginal cost affects the payment choice only through its effect on the markup.*

Proof. See appendix D.2. ■

Proposition 4 states that a decline in marginal costs (increase in productivity) leads to an increase in the markup and, thereby, to more trade credit provision. First, the linear demand creates a link where lower costs cause higher markups. Second, the financial cost mechanism developed here then links the increase in markups to an increase in the provision of trade credit. Importantly, there is no direct effect of marginal costs on the payment choice, as marginal costs do not directly affect the sign of the profit difference between trade credit and cash in advance but only affect it indirectly through their effects on markups.

2.4.2 Partial Pre-Payment

The seller could ask for a partial pre-payment; that is, the buyer pays a fraction of the overall price in advance and pays the remainder after delivery. Interestingly, partial pre-payments only represent a very small fraction of transactions in our data as well as in other data that we are aware of (e.g. Antràs and Foley, 2015). As appendix D.3 shows, the only partial pre-payment that can be optimal is a prepayment that equals production costs C . For that case, optimal profits of the seller can be derived as:

$$\Pi_S^{PP} = R - (1 + r_b^*)C. \quad (15)$$

Profits with a partial pre-payment are very similar to those with trade credit. The only difference is that now production costs are pre-paid by the buyer, and therefore the buyer's borrowing rate, r_b^* , replaces the seller's borrowing rate, r_b , in the profit expression. These profits imply the following proposition:

Proposition 5 (Trade Credit and Partial Pre-Payments)

Suppose $\mu > 1 + r_b$ and $\mu > 1 + r_b^*$. Then:

- i) If $r_b^* > r_b$, the seller pays after delivery (trade credit).
- ii) If $r_d < r_b^* < r_b$, the buyer pre-pays production costs (partial pre-payment).
- iii) If $r_d > r_b^*$, the seller pays before delivery (cash in advance).
- iv) If $r_b = r_b^*$, the seller is indifferent between trade credit and a partial pre-payment of C .

Proof. See appendix D.3. ■

The proposition is quite intuitive. Suppose borrowing costs for both firms exceed the deposit rate. Then, if the seller has lower borrowing costs, she provides full trade credit. If the buyer has a lower borrowing cost, a partial pre-payment becomes optimal to shift financing to the buyer. However, this only happens up to a point. Specifically, the buyer only pre-pays production cost. Paying more in advance would create unnecessary financing costs as any surplus funds get deposited into the bank by the seller, which only generates return r_d . Importantly, under partial pre-payment, the seller still extends some trade credit to the buyer. Thus, the possibility of partial prepayment does not eliminate the financing cost advantage of trade credit. Full cash in advance only becomes optimal when the borrowing rate of the buyer is so low that it is below the deposit rate of the seller. If the borrowing rates are the same for the buyer and the seller, the seller is indifferent between trade credit and a partial pre-payment.

Why are partial pre-payments not more common? One possible explanation that is beyond the scope of this model are legal frictions. Specifically, partial pre-payments can be problematic from a legal perspective, as at any point in time, the legal ownership has to be assigned to one of the two parties. A buyer may be reluctant to pre-pay a fraction of the price without obtaining legal ownership. Conversely, the seller may be hesitant to transfer ownership rights before receiving the full payment. Adding this friction would be a promising extension to our model.¹¹

2.4.3 Bargaining between Buyers and Sellers

So far, we derived results assuming that the seller has all the bargaining power. To generalize the results, we extend the model to allow for different bargaining weights for the buyer and the

¹¹Another way to make partial pre-payments less desirable would be to build on [Burkart and Ellingsen \(2004\)](#) and assume that cash is more easily diverted than goods.

seller, assuming that firms Nash-bargain over the surplus. We derive the following corollary, with details in appendix D.4:

Corollary 2 (Payment Choice and Bargaining Power)

Suppose the seller has some bargaining power ($\theta \in (0, 1]$). Then all predictions in propositions 1 and 2 hold for the case where both firms have bargaining power.

Proof. See Appendix D.4. ■

The corollary states that introducing bargaining power for both sellers and buyers does not affect our main results on trade credit and markups. In particular, all predictions in proposition 1 and proposition 2 continue to hold, as long as the seller has some bargaining power. This is because the seller can still charge a positive markup over marginal costs to the buyer and the financing cost advantage of trade credit remains active.

2.4.4 Letters of Credit

Letters of credit are a payment form used exclusively in international trade transactions. With a letter of credit, banks serve as intermediaries in the transaction to resolve the two-sided commitment problem between the buyer and the seller. The buyer pays a fee to the bank and commits to paying the seller.¹² The seller only receives payment from the bank after providing proof of shipment or delivery. Under these assumptions on commitment and letters of credit, the following proposition can be derived, with details in appendix D.5:

Proposition 6 (Payment Choice: with Commitment Problem)

The choice between trade credit and letters of credit is independent of the markup, μ .

Proof. See appendix D.5. ■

The markup does not affect this choice because both trade credit and a letter of credit require the seller to finance the production costs and with both options payment only occurs after delivery. Therefore, trade credit increases with markups only at the expense of cash in advance, which we confirm empirically in section 5.3.

¹²This commitment can either reflect a long-term relationship with the bank or may require a deposit in the bank up to the value of the letter of credit. For tractability, we assume that it is sufficient for the buyer to pay the letter of credit fee in advance.

3 Empirical Approach

This section presents the main empirical specifications, discusses threats to identification, and introduces our instrumental variable approach. In addition, it lays out the methodology we use to compute markups at the firm-product level.

3.1 Empirical Strategy

The empirical analysis looks at two outcome variables that are each constructed as trade-weighted averages at the firm-product-destination year level: (i) the share of exports sold on trade credit (extensive margin); and (ii) trade credit maturity (intensive margin).

The model predictions are tested in two steps. First, the outcome variables are regressed on firm-product markups. According to propositions 2 and 3, the coefficient on markups should be positive in this specification. Second, different versions of the following regression at the firm-product-destination-year level are estimated:

$$TC_{ijpt} = \beta_1 \ln(\mu_{ipt}) + \beta_2 \ln(\mu_{ipt}) \times r_{b,jt}^* + \gamma X_{ijpt} + \delta_i + \delta_{jt} + \delta_p + \varepsilon_{ijpt}, \quad (16)$$

where i denotes a firm, p a product, j a foreign country, and t denotes a year. TC_{ijpt} stands for the two outcome variables discussed above. μ_{ipt} is the markup, which we compute at the firm-product level following the methodology presented below in section 3.4. $r_{b,jt}^*$ denotes the borrowing rate in the destination country. This interest rate is at the country-year level and does not fully capture the buyer-specific borrowing costs in the model, which would, however, likely be endogenous. At the same time, the country-level borrowing cost should be a good proxy for the average cost of borrowing across all buyers in a destination country.

The main coefficient of interest is β_2 on the interaction term between the markup, μ_{ipt} , and the foreign borrowing rate, r_b^* . Propositions 2 and 3 predict that the financing cost advantage of trade credit increases with the product of the markup and the destination country's borrowing rate, r_b^* .¹³ That is, $\beta_2 > 0$.

¹³Propositions 2 and 3 also have predictions on the interaction between markups and the domestic deposit rate r_d . However, as we study data from only one exporting country, Chile, there is very limited variation in that variable. In appendix table G.2, we interact the markup with the difference between the foreign borrowing rate and the domestic deposit rate, and the results are basically unchanged from those in table 3. In fact, in

The baseline specification includes firm (δ_i), product (δ_p) and country-year (δ_{jt}) fixed effects. While we first present results with firm, product, and destination-year fixed effects, our preferred specification includes firm-product-year and destination-year fixed effects. We cluster standard errors at the country-by-firm level.

3.2 Identification

There are two main challenges to identification in our setup that we discuss in detail below.

Omitted Variables. Omitted variables could pose a threat for identification if they directly affect both markups and trade credit use. For instance, financially constrained firms tend to provide less trade credit than financially unconstrained firms (Biais and Gollier, 1997; Burkart and Ellingsen, 2004; Cuñat, 2007). At the same time, financial constraints could plausibly affect markups, for example, by forcing firms to liquidate their inventories at lower prices to obtain cash. Another omitted variable could be competition in the destination market. More competitive environments may cause lower markups and may also lead to more trade credit provision, as shown in Demir and Javorcik (2018). If the baseline OLS estimates also capture this competition channel, this will generate a downward bias in the OLS coefficients. The empirical strategy directly addresses the omitted variable concerns by including granular fixed effects at the firm-product-year and destination-year level. In addition, as described in more detail below, the IV strategy only exploits supply-side information to construct productivity estimates, resolving concerns that results are driven by changes to competition in the destination market.

Endogenous Markups. The model predicts that prices charged to the buyer are endogenous to the payment choice. In particular, a seller should charge a higher price when providing trade credit to pass on her borrowing costs to the buyer.¹⁴ This price effect implies a positive

columns 2, 4, 6, and 8, the firm-product-year FEs eliminate any effect of r_d and its interaction with the markup, and results are therefore identical for those columns.

¹⁴In addition, if there is a two-sided commitment problem and imperfect contract enforcement, the seller also requires compensation for bearing the risk that the buyer may not pay. Antràs and Foley (2015) provide suggestive evidence for the price effect of trade credit, looking at transaction-level data from a U.S.-based exporter of frozen and refrigerated food products. We also estimated the correlation between a trade credit dummy and the unit values in our export data and found that trade credit transactions, on average, have 3

correlation between trade credit choice and markups, biasing the OLS estimates upward. This concern is addressed by the fact that the instrument is constructed without any price or revenue information but only uses supply-side information, as we discuss in detail next.

3.3 Instrumental Variable Estimation

To address omitted variable and endogeneity concerns, we implement an IV strategy, using firm-product physical total factor productivity (TFPQ) as an instrument for markups. As the model with variable markups shows, an imperfect pass-through of higher efficiency to lower prices can generate a positive correlation between physical productivity and markups.

For the main specification (equation 16), the IV strategy works as follows. In the first stage, we predict firm-product markups and the interaction between firm-product markups and the destination country borrowing rate with estimated firm-product TFPQ and its interaction with the destination country borrowing rate. Importantly, when estimating the production function and computing TFPQ, we specify output and intermediate inputs in terms of physical units to avoid the so-called output and input price biases.¹⁵ As De Loecker and Goldberg (2014) explain, these biases lead to confounding measured productivity with markups. By specifying the production function in physical units, the estimated TFPQ only reflects supply-side production factors and does not reflect any demand conditions, which is crucial for the validity of TFPQ as an instrument for markups.

In the second stage, we regress each outcome variable, TC_{ijpt} , on predicted log markups, $\widehat{\ln(\mu_{ipt})}$, predicted interactions between markups and the destination country borrowing rate, $\widehat{\ln(\mu_{ipt}) \times r_{b,jt}^*}$, controls, and fixed effects:

$$TC_{ijpt} = \beta_1 [\widehat{\ln(\mu_{ipt})}] + \beta_2 [\widehat{\ln(\mu_{ipt}) \times r_{b,jt}^*}] + \gamma X_{ijpt} + \delta_i + \delta_{jt} + \delta_p + \varepsilon_{ijpt} \quad (17)$$

Exclusion Restriction and Identification The exclusion restriction for using TFPQ as an instrument for markups requires that conditional on the fixed effects, TFPQ (and its interaction with destination-country borrowing rates) only affects the payment choice indirectly through percent higher unit values. Note that this is a relatively small effect relative to the dispersion of log markups (standard deviation of 0.37).

¹⁵Appendix E provides technical details on the estimation of the production function at the firm-product level.

its effects on markups (and its interaction with destination-country borrowing rates).

Our framework is consistent with this restriction, as shown in proposition 4. Nevertheless, there may be factors outside the model that might link productivity directly to trade credit provision. For instance, higher efficiency may reflect better management practices (Bloom and Reenen, 2007) or may imply that firms are less financially constrained (Aghion et al., 2019). An additional concern is that sellers may be particularly efficient in producing a certain product. The higher efficiency may lead to higher bargaining power over the buyer of the product, which may indirectly affect trade credit provision. In all these cases, efficiency would be linked to an omitted variable that may affect trade credit preference at the firm-year or firm-product-year level. The firm-product-year fixed effects in our preferred specification directly control for these alternative mechanisms.

To summarize, instrumenting markups by physical productivity and including detailed fixed effects resolves the two main endogeneity concerns discussed above. The granular fixed effects address omitted variable concerns. The IV resolves concerns about changes in competition in destination markets, as we only exploit changes in markups that are due to differences in physical productivity at the firm-product level. The IV also addresses the concern that firms charge higher prices under trade credit, as the physical productivity estimate only reflects differences in technology and efficiency at the firm-product level and does not rely on revenue or price data.

3.4 Markups Estimation

To test the model, we construct markups at the firm-product-year level following the production-based approach by De Loecker et al. (2016). This methodology requires minimal working assumptions, is flexible with respect to the underlying demand system, and only requires production data. We briefly explain the main elements of this methodology and relegate a more detailed technical discussion to appendix E.

The main insight in De Loecker et al. (2016) is that price-cost markup of a firm-product can be computed as the ratio between two elements: (i) The output elasticity of product p with respect to any flexible input V (θ_{ipt}^V), and (ii) the expenditure share of the flexible input V (relative to the sales of product p ; s_{ipt}^V). The former element requires the estimation of the production function at the firm-product level, while the latter component can be directly

computed from our data. We briefly explain how we compute each of these elements next.

To estimate the production function coefficients, we specify a Cobb-Douglas production function, with labor, capital, and materials as production inputs for each product p .¹⁶ We measure output in terms of physical units and deflate materials expenditure with a firm-specific input price index. In this way, we avoid the occurrence of input and output price biases (see [De Loecker and Goldberg, 2014](#), for details, and the discussion at the end of this section). To identify the production function coefficients in multi-product firms, we follow [De Loecker et al. \(2016\)](#), and assume that products are produced with the same technology in single- and multi-product firms. Hence, we identify the production function coefficients for all firms-products using the subset of single-product firms.¹⁷ We estimate the production function coefficients following the methodology proposed by [Akerberg et al. \(2015\)](#) to control for the endogeneity of firms' inputs choice.¹⁸

The second component needed to compute markups is the expenditure share, which is observed at the firm level. To estimate this element for products within a firm, we follow [Garcia-Marin and Voigtländer \(2019\)](#) and proxy for product-specific input use, assuming that inputs are used approximately in proportion to overall variable cost shares. For this, we take advantage of the fact that ENIA provides information on total variable costs (labor cost and materials) for each product produced by the firms. Finally, we compute the expenditure share by dividing the value of material inputs by product-specific revenues, which are observed in the data.

¹⁶We consider the widely used Cobb-Douglas production function for our analysis to keep comparability with the U.S. based results, where we use the production function estimates from [De Loecker et al. \(2020\)](#).

¹⁷The main limitation of this approach is that it restricts economies of scope on the production side, but as we discuss in the robustness checks section, our main results also hold when using average product margins (directly observed in our data) or when computing markups at the firm-level, which are not subject to this criticism.

¹⁸In addition, we allow past exporting and investment decisions to affect firms' productivity, and include the probability of remaining single-product to correct for the bias that results from firm switching non-randomly from single to multi-product production (see [De Loecker et al., 2016](#), for details).

4 Data

The main analysis uses information for the universe of Chilean manufacturing exporters over the period 2003-2007. In addition, we confirm our results with company-level Compustat data from the United States. A key advantage of the Chilean data is that it provides detailed information on physical inputs and outputs, allowing better identification of the main mechanism. This section reviews the main features of the Chilean data and describes the sample. We describe the U.S. Compustat data in section 6.

4.1 Details on the Chilean Data

The Chilean data combines information from two primary data sources. The first dataset is collected by the Chilean National Customs Service and provides information for the universe of Chilean exports. The data is available for the 90 main destinations of Chilean exports, accounting for over 99.7% of the value of overall national exports in our sample period. The dataset details the exporter’s identity, the importing country, the 8-digit HS code, FOB value and volume of the merchandise, the payment due date, and the export transaction’s financing mode. This last feature is key for our purpose, as it allows us to identify if each transaction was paid in advance (cash in advance – CIA), post-shipment (trade credit – TC), or with other modes (such as letters of credit or two-part contracts). In addition, we observe the expected payment date, which we use to calculate trade credit maturities.

We complement the customs-level data with production-level data from the *Encuesta Nacional Industrial Anual* (Annual National Industrial Survey – ENIA). ENIA is collected by the Chilean National Statistical Agency (INE), and provides annual production information for the universe of Chilean manufacturing plants with 10 or more employees, according to the International Standard Industrial Classification (ISIC), revision 3. The ENIA survey data is generally considered of high quality and has been widely used in research.¹⁹ It surveys approximately 4,900 manufacturing plants per year, out of which 20% are exporters. Approximately two third of the plants in ENIA are small (less than 50 workers); medium-sized (50-150 workers) and large (more than 150 workers) plants represent 20 and 12 percent, respectively. ENIA provides standard micro-level information (e.g., sales, inputs expenditures, employment, investment), and detailed

¹⁹Examples of studies using the ENIA dataset include [Pavcnik \(2002\)](#), [Levinsohn and Petrin \(2003\)](#), [Akerberg et al. \(2015\)](#), [Garcia-Marin and Voigtländer \(2019\)](#), [Gandhi et al. \(2020\)](#), among others

information for each good produced (sales value, production cost, number of units produced and sold), and inputs purchased by the firm (value and volume for each input purchased by the plant). Output and input products are defined according to the Central Product Classification (CPC) at the 8-digit level, identifying 1,190 products over 2003-2007.²⁰

4.2 Sample Selection and Data Consistency

Our main analysis considers data for the sample of manufacturing exports for which the production data in ENIA is available. In the following, we explain the matching procedure between both datasets and the procedures we apply to ensure a consistent dataset. A detailed discussion of these issues is relegated to internet appendix F.

The matching procedure between the ENIA and customs datasets consists of two stages. First, we match firms using a common tax-identifier available in both datasets, allowing us to match all direct exporters in ENIA to customs. Next, we proceed to match products in ENIA and customs. For this, we first use the United Nations’ correspondence tables between CPC and HS product classifications, leading to a unique match for about 60% of the exported products in ENIA. In cases where we find multiple matches, we check the potential matches and manually assign products with concordance within 4-digit HS categories in both datasets. We drop firm-products where there is no clear connection between the product categories in both datasets and matches where the firms appear exporting non-manufacturing products. This procedure allows matching 86% of the exported firm-product observations in ENIA to customs.²¹

To ensure a consistent dataset, we follow several steps, including the deletion of observations that have missing, zero, or implausible variation in the values of any of the main variables. In the empirical analysis, we aggregate the transaction data at the annual frequency, the frequency at which we estimate markups. The final dataset consists of 88,546 firm-product-destinations-year observations. The resulting dataset represents 69.2% of the value of (non-copper) exports over the period 2003-2007.

²⁰For example, CPC disaggregates the wine industry (ISIC 3132) into 4 different categories: “Sparkling wine”, “Wine of fresh grapes”, “Cider”, and “Mosto”.

²¹The imperfect match between ENIA and customs may be because firms in ENIA report that a product is exported independently if the firms exported directly or through intermediaries. Only the former case has a match with the customs data, while the latter does not.

4.3 Additional Data

To complement the manufacturing survey and customs data, we collect information for destination countries' lending rates, as well as for destination countries' domestic inflation rates from the International Monetary Fund's *International Financial Statistics*. We use this data to construct real (ex-post) interest rates as the difference between the nominal rates and the realized inflation in the respective year.²²

4.4 Descriptive Statistics

Table 1 reports descriptive statistics for the characteristics of the firms and the products they export over the period 2003-2007. The average firm in the sample has \$22 million in total value of shipments ("exports"), employs approximately 275 workers, and export 7.5 different 8-digit HS products. Firms in the sample are larger than the average firm in the Chilean ENIA, which is expected because our sample only includes exporters.²³ Large firms (more than 150 employees) account for 44% of the firms, while small firms (less than 50 employees) only account for 23% of the firms in the sample.

The second panel in table 1 provides information disaggregated at the firm-product level. Firms export their products to an average of 3.4 different destinations. The average markup over marginal cost is 15.6%, which is slightly smaller than the average markup computed over all products in ENIA – that is, including also goods produced for the domestic market.²⁴

Focusing on payment terms, the data shows a clear dominance of trade credit. Approximately 83% of the transactions are paid for this way (third panel in table 1). About 9 percent of the transactions are paid cash in advance, and another 6 percent use letters of credit. Other contracts – including two-part contracts where a fraction of the value is paid upfront and the rest once the good arrives at the destination – are relatively rare and account for less than 1.5%

²²We drop destination-year pairs with extreme values of the real ex-post borrowing rates to avoid issues arising from very negative ex-post real rates. In practice, this correction exclude from the main analysis destination-year pairs with real ex-post borrowing rates above 35%, and below -4%.

²³A large literature documents that exporters are larger in terms of employment and sales, are more productive, and pay higher wages, among other characteristics. See for example [Bernard and Jensen \(1999\)](#) and [Bernard and Wagner \(1997\)](#).

²⁴Table F.6 in the internet appendix provides summary statistics for markups in the full ENIA dataset, aggregated at the 2-digit level. The average estimated markup is 1.3, while the median is 1.1.

of all transactions. The average maturity of trade credit is 170 days, which is notably longer than what is typically documented for domestic transactions. This difference is in large part due to the longer transport times in international trade, as the geographical distance from Chile is a strong predictor of trade credit maturity in our data (figure B.2). Finally, the last panel in table 1 provides summary statistics on country-level variables that we use in the empirical analysis.

Table 1. Summary Statistics

	Mean	Std. Dev.	P25	P50	P75	Obs.
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Firm Characteristics</i>						
Export Value (US\$)	22,097.8	226,983.9	163.8	1,005.4	5,560.5	3,381
Employment	274.6	527.1	53	120	284	3,381
# Exported products (at the 8-digit HS level)	7.5	9.5	2	5	10	3,381
<i>Firm-product Characteristics</i>						
# Destinations by firm-product-year	3.4	5.1	1	1	4	25,444
Markups (in logs)	0.156	0.371	-0.119	0.111	0.383	25,444
Physical total factor productivity (in logs)	0.386	3.335	-2.567	1.305	2.991	25,444
<i>Firm-product-destination Characteristics</i>						
Trade-Credit Share	83.1	35.3	100	100	100	88,546
Cash-in-Advance Share	9.3	27.5	0	0	0	88,546
Letters-of-Credit Share	6.3	22.6	0	0	0	88,546
Trade Credit Maturity (days)	169.5	94.6	90	168	270	77,328
<i>Country Characteristics</i>						
Foreign borrowing rate	5.162	4.226	2.547	4.372	6.762	353
Chilean deposit rate	0.899	0.559	0.853	0.873	1.152	353

Notes: The table lists the summary statistics for the variables used in the paper's baseline analysis sample. It comprises customs-level data for the universe of Chilean manufacturing exporters that can be matched to the Chilean Annual Manufacturing Survey (ENIA), over the period 2003-2007.

5 Results

Before turning to the econometric evidence, we illustrate our main results in figure 3. The figure shows four binscatter plots. Panel A shows charts for trade credit shares (extensive margin), while panel B shows charts for trade credit maturities (intensive margin). In all charts, the average value of the outcome variable in each bin is plotted against the average firm-product markup (in logarithm). For all variables, the plot is based on residuals after taking out country-year fixed effects. Charts on the left show data for countries with borrowing rates that are above the median rate across years and destinations, while charts on the right show data for

countries where borrowing rates are below the median. As predicted by proposition 2, for high-interest-rate destinations, there is a clear positive relationship between the intensive and extensive margins of trade credit and markups. In contrast, for low-interest-rate destinations, the relationship is either weaker (right chart, Panel A) or flat (right chart, Panel B).²⁵

The initial graphical evidence is therefore fully consistent with our model predictions: the financing cost advantage of trade credit should increase with the markup, especially when buyers face high borrowing costs.

5.1 Main Results

We now turn to the econometric evidence.

Trade credit use increases in the markup. Table 2 presents our baseline results on trade credit use and the level of markups. Columns 1 through 3 show results for the trade credit share, while columns 4 through 6 report results on trade credit maturities.

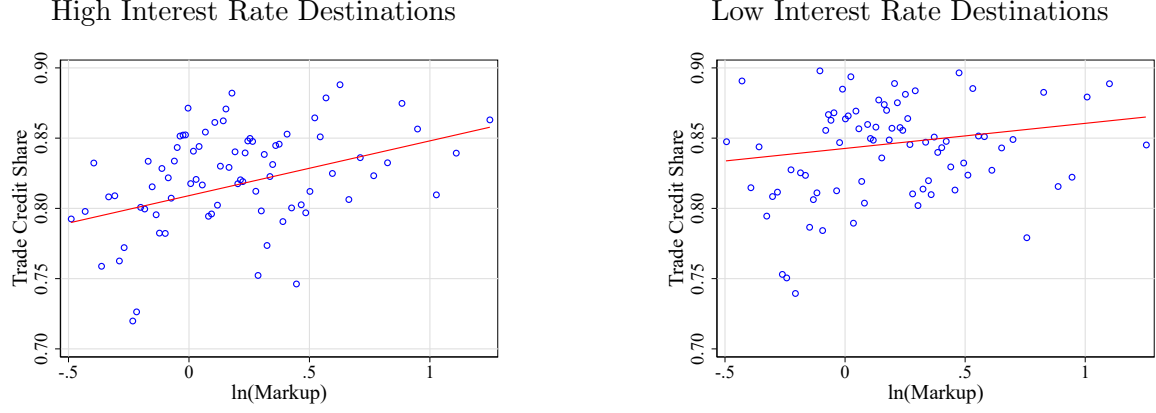
Columns 1 and 4 report OLS results. In line with propositions 2 and 3, and the evidence presented in figure 3, we find a positive and highly significant coefficient for markups both for the extensive margin and the intensive margin. Columns 2 and 5 show the first stage regressions, where we instrument markups by TFPQ. The first stages work well, with F-statistics substantially above the Stock-Yogo critical value of 16.4 for 10% maximal IV bias. Consistent with proposition 4, the coefficients on TFPQ are positive and highly significant, implying that firms charge higher markups in products they produce more efficiently. The magnitude of the first-stage coefficient in column 2 implies that a ten percent increase in TFPQ is associated with an increase in markups of 0.5%. Columns 3 and 6 present the instrumental variable results. The estimated coefficients are positive and highly significant at the 1% level. The coefficients are also notably larger than the OLS coefficients in columns 1 and 4, indicating that results are biased towards zero when not instrumenting for markups.

Effects are economically meaningful. Based on the IV coefficients in columns 3 and 6, an increase of one standard deviation in the firm-product log markup (0.371 log points) increases

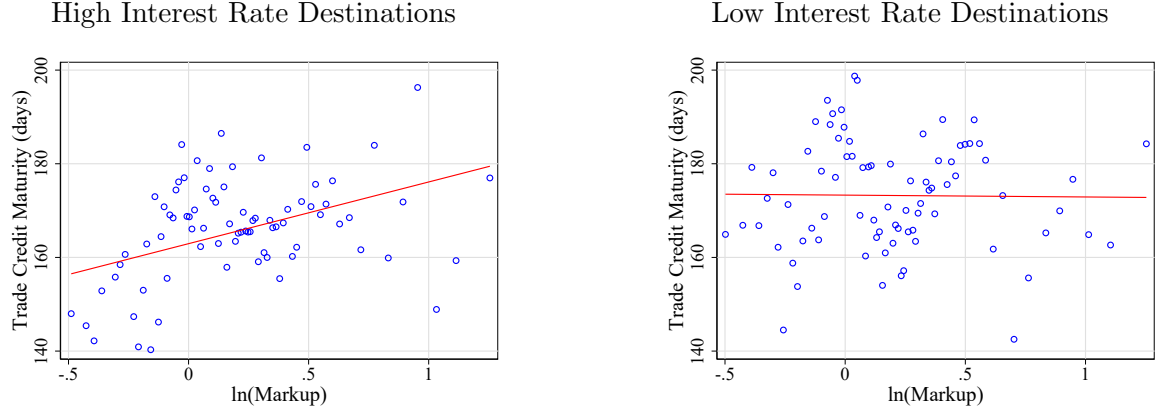
²⁵Figure B.3 replicates the extensive margin of figure 3 (panel A) for the share of transactions financed through cash in advance and letters of credit contracts. These figures suggest that firms increase trade credit use with markups at the expense of cash in advance contracts. The use of letters of credit contracts, in contrast, appears unresponsive to markups.

Figure 3. Trade Credit Share, Markups and Interest Rates

A. Trade Credit Share (extensive margin)



B. Trade Credit Maturity (in days, intensive margin)



Notes: The figures show binscatter plots of the trade credit share (panel A) and trade credit maturity (panel B) against markups (in logs). Each panel splits the data for export destinations with borrowing rates above and below the median rate across destinations. Trade credit share and trade credit maturity are computed at the firm-product-destination level, and markups are computed at the firm-product level, following the methodology by [De Loecker et al. \(2016\)](#). All figures control for destination-year fixed effects.

the trade credit share by 4.1 percentage points and lengthens the average trade credit maturity by 6.8 days. Combining the two margins implies an increase in trade credit of 12.6 days, with each margin explaining about half of the effect.²⁶

²⁶To compute the total effect, first note that the average total trade credit length across all transactions equals the product between the trade credit share and trade credit maturity. Then, the implied effect of markups on total trade credit equals the extensive margin effect (0.041) times the average trade credit maturity (169.5 days) plus the intensive margin effect (6.8 days) times the average trade credit share (0.832).

Table 2. Trade Credit and Firm-Product Markup: Baseline Regressions

Dependent Variable:	Trade Credit Share			Trade Credit Maturity		
Specification:	OLS	FS	IV	OLS	FS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
ln(Markup)	2.208*** (0.491)	—	11.07*** (2.971)	5.347*** (1.940)	—	18.331** (7.276)
ln(TFPQ)	—	0.051*** (0.0038)	—	—	0.055*** (0.0043)	—
First Stage F-Statistic	—	183.5	—	—	154.7	—
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Product FE	Yes	Yes	Yes	Yes	Yes	Yes
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	88,546	88,546	88,546	77,328	77,328	77,328

Notes: All regressions are run at the firm-product-destination level (with products defined at the HS8 level). Trade credit shares are computed as the ratio of the FOB value of trade credit transactions to the FOB value of all export transactions over a year. Trade credit maturity corresponds to the days from shipping to the agreed payment due date in the trade credit contract. Markups and TFPQ are computed at the firm-product level. Columns 1 and 4 report OLS estimates. The first stage results of the IV regressions are reported in columns 2 and 5, together with the (cluster-robust) Kleibergen-Paap rKWald F-statistic. The corresponding Stock-Yogo value for 10% maximal IV bias is 16.4. IV results are reported in columns 3 and 6. Standard errors (in parentheses) are clustered at the firm-product level. Key: *** significant at 1%; ** 5%; * 10%.

Markups effects increase in the borrowing rate abroad. Next, we present results on the interactions between markups, trade credit, and interest rates, based on equation (16). Columns 1, 2, 5, and 6 of table 3 report OLS estimates and columns 3, 4, 7, and 8 report IV estimates. In all specifications, standard errors are clustered at the firm-destination level. All first stage regressions have F-statistics clearly above the 16.4 threshold (see table G.1 for the full first stage regressions).

Columns 1 and 5 estimate the baseline specification, including firm, product, and destination-year fixed effects. The coefficient on the interaction term between the markup and the buyer's borrowing rate, r_b^* , is positive and highly significant for trade credit shares. The coefficient also has the expected sign for trade credit maturities but is insignificant. Columns 2 and 6 add firm-product-year fixed effects with little effect on estimated coefficients.

IV coefficients for the interaction term presented in columns 3, 4, 7, and 8 are notably larger than the OLS estimates. Moreover, they are highly significant both for trade credit shares and for trade credit maturities. As before, adding the more stringent firm-product-year fixed effects does not materially alter the estimates. However, these additional fixed effects tighten the identification of the interaction term between markups and destination country borrowing rate,

as they address concerns about omitted variables that may directly link physical productivity to trade credit use at the firm-product level.

Table 3. Trade Credit and Firm-Product Markup: Heterogeneity

Dependent Variable:	Trade Credit Share				Trade Credit Maturity			
Specification:	OLS		IV		OLS		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ln(Markup)	0.654 (0.950)	—	4.068 (5.161)	—	3.949 (2.733)	—	0.1425 (11.825)	—
ln(Markup) $\times r_b^*$	0.282** (0.140)	.301** (0.148)	1.157** (0.588)	1.290** (0.590)	0.257 (0.301)	0.480 (0.403)	3.054** (1.537)	3.412** (1.675)
First Stage F-Statistic	—	—	172.5.0	49.5	—	—	145.3	44.7
Firm FE	Yes	No	Yes	No	Yes	No	Yes	No
Product FE	Yes	No	Yes	No	Yes	No	Yes	No
Destination-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-product-year FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	88,546	88,546	88,546	88,546	77,328	77,328	77,328	77,328

Notes: The table reports the coefficient estimates from equation (16). All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Trade credit share (columns 1 to 4) corresponds to the ratio of the FOB value of trade credit transactions to the FOB value of all export transactions over a year. Trade credit maturity (columns 5 to 8) corresponds to the days from shipping to the agreed payment due date in the trade credit contract. Markups are computed at the firm-product level (products are defined at the 5-digit CPC level). Columns 1, 2, 5, and 6 report OLS, while columns 3, 4, 7, and 8 report IV results using TFPQ (and its interaction with the interest rate spread) as an instrument for markups (and its interaction with the interest rate spread). All IV regressions report the (cluster-robust) Kleibergen-Paap rK Wald F-statistic; the corresponding Stock-Yogo value for 10% maximal IV bias is 16.4. All regressions control for the logarithm of firm employment. Standard errors (in parentheses) are clustered at the firm-destination level. Key: *** significant at 1%; ** 5%; * 10%.

Estimated effects for the interaction terms are also economically relevant. Consider two firm-products at the 25th (markup of 0.89) and 75th percentile (markup of 1.47) of the markup distribution, respectively. Based on the coefficient in column 8, a one-standard-deviation higher borrowing rate (4.2 percentage points) in the destination country increases the share of trade credit by 3.1 percentage points and the average trade credit maturity by 10.7 days for a firm with a markup at the 75th percentile relative to a firm with a markup at the 25th percentile. This implies that total credit increases by 14.2 days more for a firm at the 75th percentile than for a firm at the 25th percentile, with the intensive margin accounting for 63 percent of this effect.

To summarize, both the baseline regressions and the interaction term regressions generate results fully in line with the model predictions in propositions 2 and 3. That is, the financing cost advantage of trade credit increases with the markup and its interaction with destination country borrowing rate, increasing trade credit use both at the extensive and intensive margin.

5.2 Robustness Checks.

In this subsection, we discuss the most important robustness checks, with additional robustness checks reported in appendix G.

Average product margins. We begin by studying whether our results depend on the method used to estimate markups. Our baseline markup measure is derived following De Loecker et al. (2016). ENIA provides information for product-level price-cost margins, an alternative proxy for markups that does not depend on a particular methodology. In particular, ENIA reports the variable production cost per product, defined as the sum of raw material and direct labor costs involved in the production of each product. Product margins can be derived by dividing prices (unit values) over this reported measure of average variable cost.²⁷ Columns 1-4 of table A.2 estimate our IV regressions for the trade credit share and maturity, respectively, when using margins as a proxy for markups. The regressions deliver qualitatively similar results with highly significant coefficient estimates.

Firm-level markups. Our baseline markup measure uses reported average variable cost shares to allocate inputs to outputs. We do not need to assume this when computing markups at the firm level. Results in table A.2 show similar point estimates when using the firm-level markups instead of firm-product-level markups.

Additional Fixed Effects. While our main analysis with interaction terms allows for a very rich set of fixed effects, we can also expand the set of fixed effects for the baseline estimation (table 2), as shown in table A.3. In the table, all coefficients are highly significant except for that in column (6), which is, however, only marginally insignificant and has a size that is broadly in line with the previous estimates.²⁸

²⁷Note that the average variable cost is self-reported by managers, making the application of rules of thumb likely. Nevertheless, as Figure G.1 in the appendix shows, there is a remarkable positive relationship between markups and reported margins, suggesting that our markup estimates yield sensible information about the profitability of the products produced by the firm.

²⁸Adding firm-year fixed effects weakens the exclusion restriction notably, as differences in access to finance, the quality of management or other factors that vary at the firm-year level and that may affect both markups and trade credit are now controlled for. When we include firm-year fixed effects, identification comes from variation in markups across products within the same firm. Thus, any violation of the exclusion restriction

To summarize, our baseline results are robust to using alternative markup measures, estimating markups at the firm level, and to the inclusion of more stringent fixed effects.

5.3 Trade Credit, Cash in Advance, and Letters of Credit

In our baseline model, we study the trade-off between trade credit and cash in advance. However, as discussed earlier, in international trade, there is a third payment form called letters of credit that is used quite frequently. In our data, it has a share of 6.3 percent compared to 83 percent for trade credit and 9.3 percent for cash-in-advance (see table 1). In this subsection, we provide evidence on the predictions from our extended model with letters of credit, showing that markups affect the choice between trade credit and cash in advance, but not that between trade credit and letters of credit.

According to the model, buyer-seller pairs substitute cash in advance with trade credit as the markup increases, and this effect should be stronger in destinations with higher borrowing rates (proposition 2). In contrast, the choice between trade credit and letters of credit should be independent of the markup (proposition 6).

These predictions align with the binscatter plots in figure B.3. Panel A show that the use of cash in advance declines in markups, with the effect being stronger for destinations with relatively high borrowing rates. The charts are almost the exact mirror image of panel A in figure 3, suggesting that firms with a higher markup increase their use of trade credit at the expense of cash in advance. Letters of credit use, in contrast, is relatively unresponsive to markups, both in high and low-interest rate destinations (panel B).

Table A.4 confirms these results econometrically, focusing on the interaction-term specifications from table 3. Column 1 repeats the baseline using trade credit share as the dependent variable for reference. Then, in columns 2 and 3, we change the dependent variables, using the share of FOB export value financed through cash in advance or letters of credit. As predicted by the theory, the coefficients for the cash in advance share (column 2) closely mirror those for trade credit (column 1), suggesting that firms substitute cash in advance with trade credit in destinations with higher borrowing costs. In contrast, the letter of credit share appears unresponsive to the interaction terms between the markup and the borrowing rate.

Finally, in columns 4 and 5, we study the source of variation driving the coefficient in

would need to operate at the firm-product-year level.

column 1, dropping transactions financed exclusively using letters of credit (column 4) or cash in advance (column 5). Results confirm the conclusions drawn from columns 2 and 3, suggesting that most of the variation that explains the financing cost advantage of trade credit comes from firms substituting between cash in advance and trade credit. When dropping the transactions financed with letters of credit, we obtain a very similar coefficient to our baseline estimate in column 1. In contrast, when dropping cash in advance transactions, we obtain a non-significant coefficient for the interaction between the markup and the borrowing rate.

6 Evidence from the United States

In this section, we repeat the main empirical analysis using firm-level data from the United States for the period 1965-2016 from Compustat. This dataset has been used extensively across different fields (more recently in [De Loecker et al., 2020](#), who document the evolution of market power in the United States). Compustat samples relatively few U.S. companies each year. However, these companies tend to be large and account for a large share of private sector employment and sales.

In the Compustat data, we calculate trade credit use as the ratio of accounts receivables over sales. Account receivables are the total value of trade credit outstanding and therefore reflect both the extensive and the intensive margins of trade credit. As before, markups are estimated following the methodology in [De Loecker et al. \(2016\)](#). In the computation of markups, we consider the cost of goods sold (COGS) as the relevant flexible input.²⁹ We take the elasticity of COGS with respect to output directly from [De Loecker et al. \(2020\)](#), and calculate the share of COGS in sales from the data. As for the case of the Chilean data, we exclude companies with missing or zero NAICS code, sales, or COGS, and firm-years with trade credit share above 100 percent or with extreme values for markups (below the 2nd or above the 98th percentiles of the markups distribution).

One important limitation of Compustat relative to the Chilean export data is that it does not provide information for output in terms of physical units. This prevents us from estimating physical productivity and using the instrumental variable approach that we use in the main

²⁹COGS is a composite that includes all expenditure incurred by firms in the production of the goods. While its specific composition varies across sectors, it mostly reflects variation in intermediate inputs, labor cost, and energy.

analysis.

We find very similar results in the U.S. data as in the Chilean data. As shown in figure B.4, the U.S. data also exhibit a clear positive relationship between trade credit use and markups, that seems to be even stronger than the one we found for Chile. This is confirmed in columns 1 and 2 of table A.5, that show a strong positive correlation between markups and trade credit, controlling for industry-year (at the 2-digit level) and firm fixed-effects.³⁰ In column 3, we present results on an interaction between the markup and the real (ex-post) effective Fed Funds Rate, our measure of borrowing costs in the U.S. data. Consistent with our theory and the evidence for Chile, the interaction term is positive and highly significant (again, with a similar magnitude as the OLS estimate for Chile). Altogether, the results for the United States suggest that our findings for Chile generalize to the case of large U.S. firms as measured in Compustat: Total trade credit use increases with markups, especially when borrowing is more expensive, which is consistent with the financing motive for trade credit use outlined in our model.

7 Concluding Remarks

Trade credit is the most important form of short-term finance for firms in emerging and developed economies, including the United States. This paper proposes a model where trade credit reduces borrowing from banks, which gives rise to a financial cost advantage of trade credit. In line with this financing cost advantage, Chilean firm-product-destination level data and U.S. firm-level data show that trade credit use and trade credit maturities increase with markups, and that these effects increase with the buyers' borrowing costs.

An important conceptional point of the model is that the choice that firms face is not between trade credit and bank finance, but rather whether the buyer or the seller borrows from a bank. If the seller borrows, the buyer gets trade credit. If the buyer borrows, the seller receives cash in advance. The key result of the theory is that when there are positive markups and diversion risk creates an interest rate spread, it is never optimal for the buyer to borrow and pre-pay the full invoice. Instead, the seller should either provide full trade credit to the buyer, or the buyer should only pre-pay production costs, as this minimizes bank credit and

³⁰As the data varies at the firm-year level, we only control for firm and industry-year fixed effects, and cluster standard errors at the firm level.

hence total financial intermediation costs. That is, trade credit has a financing cost advantage because this payment form minimizes the total amount of bank funds that are exposed to diversion risk.

Our model implies that firms’ payment choices affect the aggregate borrowing level, making the size of the financial sector endogenous. This prediction is qualitatively consistent with recent developments in aggregate U.S. data that suggest rising markups (as estimated by [De Loecker et al., 2020](#)) and more use of trade credit over time. As higher markups increase the financing cost advantage of trade credit, making it more attractive, firms may rely more on this financing form and less on the formal financial sector. Future work could shed more light on how heterogeneity in the adoption of trade credit may affect the size and development of the financial sector. The last point may be particularly relevant for developing and emerging economies, where financial frictions and diversion risks are more significant, and hence the potential savings from using trade credit are more prominent.

References

- Akerberg, Daniel A., Kevin Caves, and Garth Frazer**, “Identification Properties of Recent Production Function Estimators,” *Econometrica*, 2015, *83* (6), 2411–2451.
- Adelino, Manuel, Miguel A. Ferreira, Mariassunta Giannetti, and Pedro Pires**, “Trade Credit and the Transmission of Unconventional Monetary Policy,” NBER Working Paper 27077 May 2020.
- Aghion, Philippe, Antonin Bergeaud, Gilbert Cette, Rémy Lecat, and Hélène Maghin**, “Coase Lecture - The Inverted-U Relationship between Credit Access and Productivity Growth,” *Economica*, 2019, *86*, 1–31.
- Ahn, JaeBin**, “Understanding trade finance: theory and evidence from transaction-level data,” Technical Report, International Monetary Fund, Washington, DC 2014.
- , **Mary Amiti, and David E Weinstein**, “Trade finance and the great trade collapse,” *American Economic Review*, 2011, *101* (3), 298–302.

- Amberg, Niklas, Tor Jacobson, Erik Von Schedvin, and Robert Townsend**, “Curbing shocks to corporate liquidity: The role of trade credit,” *Journal of Political Economy*, 2021, 129 (1), 182–242.
- , **Tore Ellingsen, Tor Jacobson, and Erik von Schedvin**, “Trade Credit and Capital Bias,” Working Paper, Stockholm School of Economics 2020.
- Amiti, Mary and David E. Weinstein**, “Exports and Financial Shocks,” *The Quarterly Journal of Economics*, 2011, 126 (4), 1841–1877.
- Antràs, Pol**, “An Austrian Model of Global Value Chains,” Technical Report, National Bureau of Economic Research 2023.
- and **C Fritz Foley**, “Poultry in Motion: A Study of International Trade Finance Practices,” *Journal of Political Economy*, 2015, 123 (4), 853–901.
- Barrot, Jean-Noël**, “Trade credit and industry dynamics: Evidence from trucking firms,” *The Journal of Finance*, 2016, 71 (5), 1975–2016.
- Bernard, Andrew B. and J. Bradford Jensen**, “Exceptional exporter performance: cause, effect, or both?,” *Journal of International Economics*, 1999, 47 (1), 1–25.
- and **Joachim Wagner**, “Exports and success in German manufacturing,” *Review of World Economics (Weltwirtschaftliches Archiv)*, 1997, 133 (1), 134–157.
- Biais, Bruno and Christian Gollier**, “Trade Credit and Credit Rationing,” *Review of Financial Studies*, 1997, 10 (4), 903–37.
- Bloom, Nicholas and John Van Reenen**, “Measuring and Explaining Management Practices Across Firms and Countries,” *The Quarterly Journal of Economics*, 2007, 122 (4), 1351–1408.
- Brennan, Michael J, Vojislav Maksimovics, and Josef Zechner**, “Vendor financing,” *The Journal of Finance*, 1988, 43 (5), 1127–1141.
- Burkart, Mike and Tore Ellingsen**, “In-Kind Finance: A Theory of Trade Credit,” *American Economic Review*, June 2004, 94 (3), 569–590.

- Chod, Jiri, Evgeny Lyandres, and S Alex Yang**, “Trade credit and supplier competition,” *Journal of Financial Economics*, 2019, *131* (2), 484–505.
- Cuñat, Vicente**, “Trade Credit: Suppliers as Debt Collectors and Insurance Providers,” *The Review of Financial Studies*, 2007, *20* (2), 491–527.
- Daripa, Arup and Jeffrey Nilsen**, “Ensuring sales: A theory of inter-firm credit,” *American Economic Journal: Microeconomics*, 2011, *3* (1), 245–79.
- De Loecker, Jan and Pinelopi Koujianou Goldberg**, “Firm Performance in a Global Market,” *Annual Review of Economics*, 2014, *6* (1), 201–227.
- , **Jan Eeckhout, and Gabriel Unger**, “The rise of market power and the macroeconomic implications,” *The Quarterly Journal of Economics*, 2020, *135* (2), 561–644.
- , **Pinelopi K. Goldberg, Amit K. Khandelwal, and Nina Pavcnik**, “Prices, Markups and Trade Reform,” *Econometrica*, 2016, *84* (2), 445–510.
- Demir, Banu and Beata Javorcik**, “Don’t throw in the towel, throw in trade credit!,” *Journal of International Economics*, 2018, *111*, 177–189.
- Demirguc-Kunt, Asli and Vojislav Maksimovic**, “Firms as financial intermediaries - evidence from trade credit data,” Policy Research Working Paper Series 2696, The World Bank October 2001.
- Emery, Gary W**, “A pure financial explanation for trade credit,” *Journal of Financial and Quantitative Analysis*, 1984, *19* (3), 271–285.
- Fabbri, Daniela and Anna Maria C Menichini**, “Trade credit, collateral liquidation, and borrowing constraints,” *Journal of Financial Economics*, 2010, *96* (3), 413–432.
- Ferris, J Stephen**, “A transactions theory of trade credit use,” *The Quarterly Journal of Economics*, 1981, *96* (2), 243–270.
- Fischer, Christian**, “Optimal payment contracts in trade relationships,” 2020.
- Fisman, Raymond and Inessa Love**, “Trade credit, financial intermediary development, and industry growth,” *The Journal of finance*, 2003, *58* (1), 353–374.

- Gandhi, Amit, Salvador Navarro, and David A. Rivers**, “On the Identification of Gross Output Production Functions,” *Journal of Political Economy*, 2020, 128 (8), 2973–3016.
- Garcia-Marin, Alvaro and Nico Voigtländer**, “Exporting and Plant-Level Efficiency Gains: It’s in the Measure,” *Journal of Political Economy*, 2019, 127 (4), 1777–1825.
- Giannetti, Mariassunta, Nicolas Serrano-Velarde, and Emanuele Tarantino**, “Cheap trade credit and competition in downstream markets,” *Journal of Political Economy*, 2021, 129 (6), 1744–1796.
- Hardy, Bryan, Felipe E Saffie, and Ina Simonovska**, “Economic Stabilizers in Emerging Markets: The Case for Trade Credit,” 2022.
- Jacobson, Tor and Erik von Schedvin**, “Trade credit and the propagation of corporate failure: an empirical analysis,” *Econometrica*, 2015, 83 (4), 1315–1371.
- Kalemli-Ozcan, Sebnem, Se-Jik Kim, Hyun Song Shin, Bent E Sørensen, and Sevcan Yesiltas**, “Financial Shocks in Production Chains,” Technical Report 2014.
- Kim, Se-Jik and Hyun Song Shin**, “Theory of supply chains: a working capital approach,” BIS Working Paper 1070 2023.
- Leibovici, Fernando**, “Financial development and international trade,” *Journal of Political Economy*, 2021, 129 (12), 3405–3446.
- Levinsohn, James and Amil Petrin**, “Estimating Production Functions Using Inputs to Control for Unobservables,” *Review of Economic Studies*, 2003, 70 (2), 317–341.
- Manova, Kalina**, “Credit constraints, heterogeneous firms, and international trade,” *The Review of Economic Studies*, 2013, 80 (2), 711–744.
- Melitz, Marc J. and Giancarlo I. P. Ottaviano**, “Market Size, Trade, and Productivity,” *Review of Economic Studies*, 01 2008, 75 (1), 295–316.
- Mian, Shehzad L and Clifford W Smith**, “Accounts receivable management policy: theory and evidence,” *The Journal of Finance*, 1992, 47 (1), 169–200.
- Murfin, Justin and Ken Njoroge**, “The implicit costs of trade credit borrowing by large firms,” *The Review of Financial Studies*, 2015, 28 (1), 112–145.

- Niepmann, Friederike and Tim Schmidt-Eisenlohr**, “International trade, risk and the role of banks,” *Journal of International Economics*, 2017, *107*, 111–126.
- and —, “No guarantees, no trade: How banks affect export patterns,” *Journal of International Economics*, 2017, *108*, 338–350.
- Nilsen, Jeffrey H**, “Trade credit and the bank lending channel,” *Journal of Money, credit and Banking*, 2002, pp. 226–253.
- Olley, G. Steven and Ariel Pakes**, “The Dynamics of Productivity in the Telecommunications Equipment Industry,” *Econometrica*, 1996, *64* (6), 1263–1297.
- Paravisini, Daniel, Veronica Rappoport, Philipp Schnabl, and Daniel Wolfenzon**, “Dissecting the effect of credit supply on trade: Evidence from matched credit-export data,” *The Review of Economic Studies*, 2015, *82* (1), 333–359.
- Pavcnik, Nina**, “Trade Liberalization, Exit, and Productivity Improvements: Evidence from Chilean Plants,” *Review of Economic Studies*, 2002, *69* (1), 245–276.
- Petersen, Mitchell A and Raghuram G Rajan**, “Trade Credit: Theories and Evidence,” *Review of Financial Studies*, 1997, *10* (3), 661–91.
- Schmidt-Eisenlohr, Tim**, “Towards a theory of trade finance,” *Journal of International Economics*, 2013, *91* (1), 96 – 112.
- Schwartz, Robert A**, “An economic model of trade credit,” *Journal of Financial and Quantitative Analysis*, 1974, *9* (4), 643–657.
- Schwartz, Robert Alan and David K Whitcomb**, “The trade credit decision,” in J. Bickler, ed., *Handbook of Financial Economics*, 1979.
- Syverson, Chad**, “Macroeconomics and market power: Context, implications, and open questions,” *Journal of Economic Perspectives*, 2019, *33* (3), 23–43.
- Wilner, Benjamin S.**, “The Exploitation of Relationships in Financial Distress: The Case of Trade Credit,” *Journal of Finance*, February 2000, *55* (1), 153–178.
- Yang, S Alex and John R Birge**, “Trade credit, risk sharing, and inventory financing portfolios,” *Management Science*, 2018, *64* (8), 3667–3689.

Internet Appendix

Diversion Risk, Markups and the Financing Cost Advantage of Trade Credit

Alvaro Garcia-Marin Santiago Justel Tim Schmidt-Eisenlohr

A Online Tables

Table A.1. Diversion Risk and its Costs

	Cash in Advance (1)	Trade Credit (2)	Difference (1) – (2)
Firm-to-firm diversion risk	R	R	0
Bank loan diversion risk	R	$C(1 + r_b)$	$R - (1 + r_b)C$
Cost of borrowing	$r_b \left(\frac{R}{1+r_b} \right)$	$r_b C$	$r_b \left(\frac{R}{1+r_b} - C \right)$
Return on deposit	$r_d \left(\frac{R}{1+r_b} - C \right)$	0	$r_d \left(\frac{R}{1+r_b} - C \right)$
Net Costs	$r_b \left(\frac{R}{1+r_b} \right) - r_d \left(\frac{R}{1+r_b} - C \right)$	$r_b C$	$(r_b - r_d) \left(\frac{R}{1+r_b} - C \right)$

Table A.2. Robustness: Alternative Markup Proxies

Markup measure:	Average Price-Cost Margin				Firm-Level Markup			
Dep. Variable:	TC Share		TC Maturity		TC Share		TC Maturity	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
ln(Markup)	96.87*	—	227.4	—	17.02***	—	36.93***	—
	(53.56)		(165.2)		(4.683)		(11.38)	
log(Markup) $\times$ r_b^*	—	2.309**	—	5.448**	—	1.242**	—	3.547**
		(.906)		(2.279)		(.606)		(1.716)
First Stage F-Statistic	4.7	23.4	3.0	21.3	201.7	50.7	165.1	44.2
Firm FE	Yes	No	Yes	No	Yes	No	Yes	No
Firm-Product-Year FE	No	Yes	No	Yes	No	Yes	No	Yes
Product FE	Yes	No	Yes	No	Yes	No	Yes	No
Destination-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	87,435	87,435	76,371	76,371	86,377	86,377	75,498	75,498

Notes: The table replicates the baseline specifications in tables 2 and 3 when using the average price-cost margin (columns 1 to 4) and firm-level markups (columns 5 to 8). All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Trade credit shares (columns 1, 2, 5, and 6) are computed as the ratio of the FOB value of trade credit transactions to the FOB value of all export transactions over a year. Trade credit maturity (columns 3, 4, 7, and 8) corresponds to the days from shipping to the agreed payment due date in the trade credit contract. Standard errors (in parentheses) are clustered at the firm-product level. Key: *** significant at 1%; ** 5%; * 10%.

Table A.3. Robustness: Varying the Set of Fixed Effects

Dependent Variable:	Trade Credit Share			Trade Credit Maturity		
	(1)	(2)	(3)	(4)	(5)	(6)
ln(Markup)	7.540*** (2.678)	6.656** (3.203)	38.62** (15.41)	23.12*** (7.292)	22.67** (9.941)	41.70 (30.02)
First Stage F-Statistic	136.7	65.65	49.54	116.1	54.77	40.86
Firm FE	Yes	Yes	No	Yes	Yes	No
Country-Year FE	Yes	No	Yes	Yes	No	Yes
Product-Country FE	Yes	No	Yes	Yes	No	Yes
Country-Product-Year FE	No	Yes	No	No	Yes	No
Firm-Year FE	No	No	Yes	No	No	Yes
Observations	88,546	88,546	88,546	77,328	77,328	77,328

Notes: The table replicates table 2 including different set of fixed effect variables. All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Trade credit shares are computed as the ratio of the FOB value of trade credit transactions to the FOB value of all export transactions over a year. Trade credit maturity corresponds to the days from shipping to the agreed payment due date in the trade credit contract. Markups are computed at the firm-product level and use TFPQ as an instrument for markups. All regressions report the (cluster-robust) Kleibergen-Paap rKWald F-statistic; the corresponding Stock-Yogo value for 10% maximal IV bias is 16.4. Second-stage results are reported in column 6. Standard errors (in parentheses) are clustered at the firm-product level. Key: *** significant at 1%; ** 5%; * 10%.

Table A.4. Trade Credit, Cash in Advance, Letters of Credit and Firm-Product Markup

Sample:	Full sample			TC vs. CIA	TC vs. LC
	TC share	CIA share	LC share	TC share	TC share
Dep. Variable:	(1)	(2)	(3)	(4)	(5)
$\log(\text{markup}) \times r_b^*$	1.290** (0.590)	-1.720*** (0.494)	0.207 (0.301)	1.609*** (0.481)	-0.378 (0.415)
First stage F-Statistic	49.5	49.5	49.5	57.9	40.6
Firm-Product-year FE	✓	✓	✓	✓	✓
Destination-year FE	✓	✓	✓	✓	✓
Observations	88,546	88,546	88,546	79,425	76,057

Notes: This table replicates table 3 modifying the dependent variable (columns 1-3) and sample (columns 4-5). All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Trade credit (TC), cash in advance (CIA), and letters of credit (LC) shares correspond to the ratio of the FOB value of transactions financed through each payment form to the FOB value of all export transactions over a year. Markups are computed at the firm-product level (products are defined at the 5-digit CPC level). Columns 4-5 restrict the sample, dropping transactions financed through letters of credit (column 4) and cash in advance (column 5). All regressions are estimated using the interaction between TFPQ and the foreign borrowing rate as an instrument for markups and its interaction with the foreign borrowing rate; the (cluster-robust) Kleibergen-Paap rK Wald F-statistic is reported for each of them (the corresponding Stock-Yogo value for 10% maximal IV bias is 16.4). Standard errors (in parentheses) are clustered at the firm-destination level. Key: *** significant at 1%; ** 5%; * 10%.

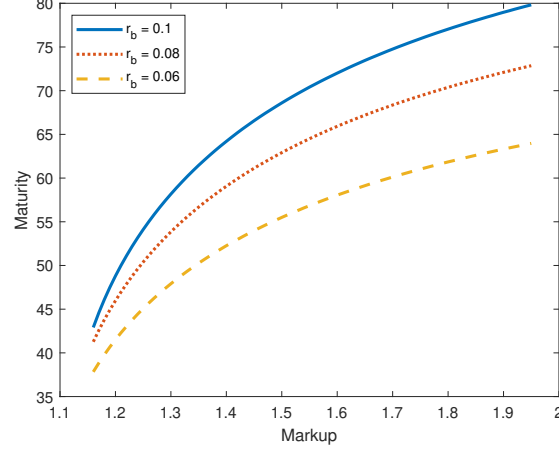
Table A.5. Trade Credit Share and Markups in the United States

	(1)	(2)	(3)
log(markup)	.0457*** (.0021)	.0227*** (.0024)	.0166*** (.0020)
log(markup) $\times$ Real Effective Fed Funds Rate	—	—	.373*** (.0053)
Industry-year FE	✓	✓	✓
Firm FE	—	✓	✓
Observations	138,680	136,789	129,125

Notes: The table estimates the main specifications using data for U.S. companies included in Compustat between 1965 and 2016. Trade credit share corresponds to the ratio of account receivables to sales. Markups are computed at the firm level using the cost of goods sold (COGS) as variable input, following [De Loecker et al. \(2020\)](#). All regressions control for the logarithm of firm employment. Standard errors (in parentheses) are clustered at the firm level. Key: *** significant at 1%; ** 5%; * 10%.

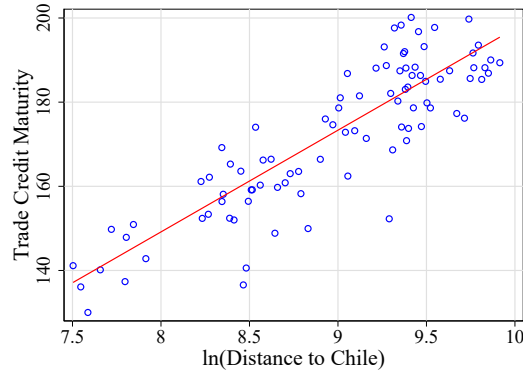
B Online Figures

Figure B.1. Trade Credit Maturity, Markups and Interest Rates



Notes: The figure illustrates the optimal maturity, t , against the markup for different borrowing rates, assuming $\tilde{\eta}(t) = 1 + (\tilde{\eta} - 1) \left(\frac{t}{T}\right)^n$, and setting $\tilde{\eta} = 0.98$, $r_d = 0.02$, $n = 2$, and $T = 180$.

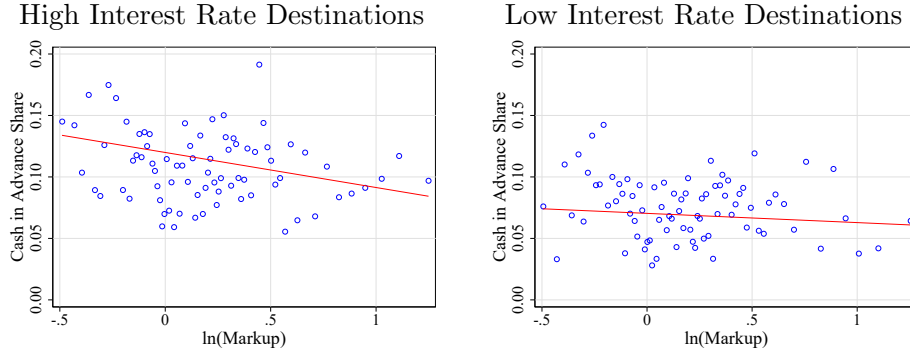
Figure B.2. Trade Credit Maturity and Distance to Chile



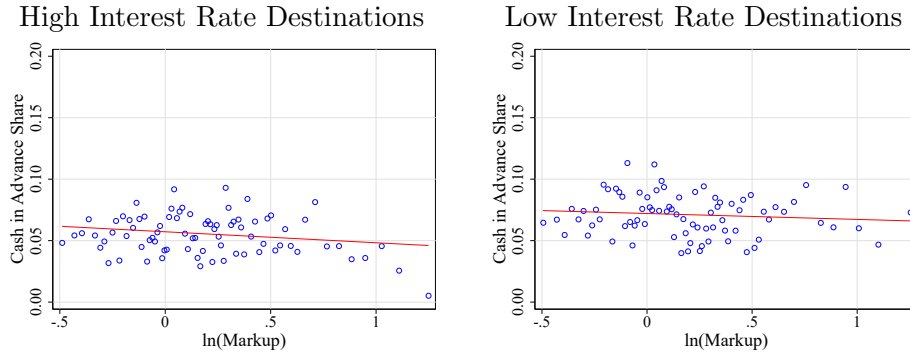
Notes: The figure shows a binscatter plot of trade credit maturity (in days) against distance to Chile (in logs). The figure controls for year fixed effects. The figure excludes Paraguay, Argentina, and Uruguay for whom geographic distance is a poor proxy for shipping time.

Figure B.3. Markups, Cash in Advance, and Letters of Credit

A. Cash-in Advance Share and Markups

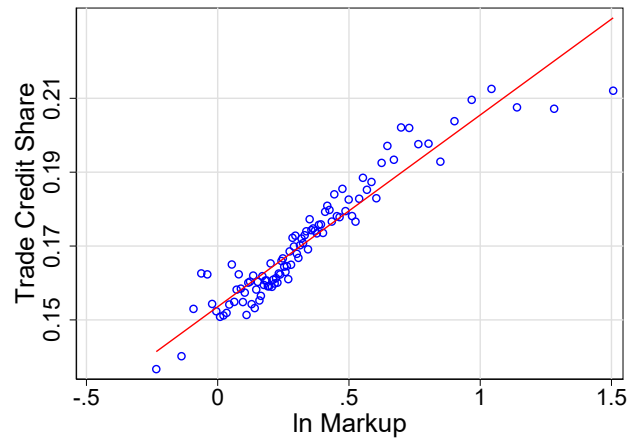


B. Letters of Credit Share and Markups



Notes: The figure shows binscatter plots of the cash in advance (panel A) and the letter of credit shares (panel B) against firm-product markups (in logs), computed as in [De Loecker et al. \(2016\)](#). In each panel, charts on the left show data for countries with borrowing rates that are above the median rate across years and destinations, while charts on the right show data for countries where borrowing rates are below the median, respectively. All figures control for destination-year fixed effects.

Figure B.4. Trade Credit Increases with Markups: U.S. Evidence



Notes: The figure shows a binscatter diagram where the average trade credit share in each bin is plotted against markups. Markups are computed at the firm level as in [De Loecker et al. \(2020\)](#), using Compustat data for 1965-2016. Markups are in terms of natural logarithms. The figure controls for 2-digit industry-year fixed effects.

C Derivation of conditions for pooling

In the following, we derive two results. First, we show that banks always offer a pooling contract that is acceptable to both types of firms. Second, we show that for sufficiently large shares of reliable firms, η and η^* , the only contracts that are used are those that are acceptable to both types of firms.

C.1 Pooling and separating cases for bank lending

When banks lend out funds, they have two choices. First, offer a rate that is only accepted by unreliable firms. Second, offer a rate that is accepted by both firms.

Lemma 1

The only equilibrium is where banks offer a contract that is accepted by both types.

Proof. There are three cases to consider. First, banks could offer a contract aimed at reliable firms only. However, unreliable firms would always accept this contract as well, as their expected payoff is strictly higher as they may divert funds, making this contract infeasible. Second, there could be a pooling contract. In the pooling case, perfect competition in the banking sector drives the borrowing rate to $1 + r_b(\tilde{\eta}) = \frac{1+r_d}{\tilde{\eta}}$. Finally, banks could offer a contract aimed at unreliable firms only. Then, perfect competition drives the borrowing rate to $1 + r_b^{SB}(\phi) = \frac{1+r_d}{\phi}$. As we assume that trade is profitable in the pooling case, that is $\frac{R}{C} = \mu > \frac{1+r_b(\tilde{\eta})}{\tilde{\eta}} = \frac{1+r_d}{\tilde{\eta}^2}$, there exists an interest rate $1 + \tilde{r}_b = 1 + r_b(\tilde{\eta}) + \epsilon$ that is acceptable to both types of firms and generates strictly positive profits for banks. As $\tilde{r}_b < r_b^{SB}(\phi)$, both types of firms would prefer this contract, which eliminates the separating contract for bad types. Therefore, the only equilibrium contract is the pooling contract where $1 + r_b = 1 + r_b(\tilde{\eta}) = \frac{1+r_d}{\tilde{\eta}}$. ■

C.2 Pooling and separating for firm contract choice

This section derives conditions for the pooling case in the model with endogenous financing costs and a two-sided commitment problem. Specifically, it derives conditions under which it is optimal for unreliable firms to imitate reliable firms and for sellers to offer terms that both types of buyers accept. In particular, we need to derive conditions to exclude the following four cases:

1. The seller asks for a payment that is only accepted by unreliable buyers under trade credit.
2. The reliable seller chooses cash in advance, but the unreliable seller chooses trade credit.
3. The seller asks for a payment that is only accepted by unreliable buyers under cash in advance.
4. The reliable seller chooses trade credit, but the unreliable seller chooses cash in advance.

Trade Credit - pooling case This is the baseline case discussed in the main text. The reliable seller maximizes:

$$\begin{aligned} \mathbb{E}[\Pi_{RS}^{TC,P}] &= \tilde{\eta}^* P^{TC,P} - (1 + r_b(\tilde{\eta}))C, \\ \text{s.t. } \mathbb{E}[\Pi_{RB}^{TC,P}] &= R - P^{TC,P} \geq 0, \end{aligned}$$

and chooses $P^{TC,P} = R$. This implies the following expected profits for reliable and unreliable sellers under pooling, respectively:

$$\begin{aligned} \mathbb{E}[\Pi_{RS}^{TC,P}] &= \tilde{\eta}^* R - (1 + r_b(\tilde{\eta}))C \\ \mathbb{E}[\Pi_{US}^{TC,P}] &= \tilde{\eta}^* R - \phi(1 + r_b(\tilde{\eta}))C, \end{aligned} \tag{C.1}$$

where unreliable sellers have higher expected profits, as there is a chance that they can divert bank funds, so they only repay with probability ϕ .

Trade Credit, Separating Case 1 *The seller asks for a payment that is only accepted by unreliable buyers under trade credit.*

Then, the payment exceeds revenues, $P^{TC,S} > R$. Unreliable buyers still accept this contract, as they know that they can deviate with probability ϕ . Expected profits of an unreliable buyer under separation are:

$$E[\Pi_{UB}^{TC,S1}] = R - \phi P^{TC,S1}.$$

In this case, the seller picks $P^{TC,S1} = \frac{R}{\phi}$. Importantly, reliable buyers now reject the contract, so that the exporter only gets the initial contract accepted with probability $1 - \eta^*$, the share of unreliable firms. Expected profits of a reliable and unreliable seller under a buyer-separating contract case 1 are hence:

$$\begin{aligned} E[\Pi_{RS}^{TC,S1}] &= (1 - \eta^*)(R - (1 + r_b(\tilde{\eta}))C). \\ E[\Pi_{US}^{TC,S1}] &= (1 - \eta^*)(R - \phi(1 + r_b(\tilde{\eta}))C). \end{aligned} \tag{C.2}$$

Combining equations (C.1) and (C.2), and substituting in the pooling equilibrium borrowing rate, a reliable seller picks the pooling case as long as:

$$E[\Pi_{RS}^{TC,P}] \geq E[\Pi_{RS}^{TC,S}] \Leftrightarrow (\eta^* - (1 - \eta^*)(1 - \phi)) R \geq \eta^*(1 + r_b(\tilde{\eta}))C.$$

Which can be rearranged to:

$$\frac{R}{C} \geq \frac{\eta^*(1 + r_b(\tilde{\eta}))}{(\eta^* - (1 - \eta^*)(1 - \phi))} \tag{C.3}$$

Note that for the unreliable seller the corresponding condition is always weaker and reads:

$$E[\Pi_{US}^{TC,P}] \geq E[\Pi_{US}^{TC,S}] \Leftrightarrow (\eta^* - (1 - \eta^*)(1 - \phi)) R \geq \eta^* \phi (1 + r_b(\tilde{\eta})) C.$$

That is, if a reliable seller prefers the pooling case, an unreliable seller will also prefer this contract.

Trade Credit, Separating Case 2 *The reliable seller chooses cash in advance, but the unreliable seller chooses trade credit.* If a firm asks for a trade credit loan in this case, the bank knows it is matched with an unreliable firm, and it charges a rate to offset the risk of diversion, that is $1 + r_b^S(\phi) = \frac{1+r_d}{\phi}$. Then, expected profits of an unreliable seller are:

$$\Pi_{US}^{TC,S2} = \tilde{\eta}^* R - (1 + r_d) C \quad (C.4)$$

In equilibrium, the bank can't make a loss, so net the seller can't steal anything, so they pay $1 + r_d = \phi(1 + r_b^S(\phi))$ in expectation. To rule out case 2, we need to combine the following four profit expressions. Profits of an unreliable seller under cash in advance:

$$\Pi_{US}^{CIA,P} = (1 + r_d) \left(\frac{\tilde{\eta} R}{1 + r_b^*(\tilde{\eta}^*)} - \phi C \right)$$

Profits of a reliable seller under cash in advance:

$$\Pi_{RS}^{CIA,P} = (1 + r_d) \left(\frac{\tilde{\eta} R}{1 + r_b^*(\tilde{\eta}^*)} - C \right)$$

Profits of an unreliable seller under trade credit separating case 2:

$$\Pi_{US}^{TC,SC2} = \tilde{\eta}^* R - (1 + r_d) C$$

And profits of a reliable seller under trade credit with pooling:

$$\Pi_{RS}^{TC,P} = \tilde{\eta}^* R - (1 + r_b(\tilde{\eta}))C$$

A sufficient condition for the separating case to be dominated is that:

$$\Pi_{US}^{CIA,P} - \Pi_{US}^{TC,SC2} \geq \Pi_{RS}^{CIA,P} - \Pi_{RS}^{TC,P}.$$

If this condition holds, then the reliable seller choosing CIA implies the unreliable seller choosing CIA as well. Rearranging delivers:

$$\Pi_{US}^{CIA,P} - \Pi_{RS}^{CIA,P} \geq \Pi_{US}^{TC,SC2} - \Pi_{RS}^{TC,P}$$

Plugging in from above and simplifying delivers:

$$\tilde{\eta} \geq \frac{1}{2 - \phi}$$

We can rewrite to:

$$\eta \geq \frac{1 - \phi}{2 - \phi} \tag{C.5}$$

Cash in Advance, Separating Case 3 *The seller asks for a payment that is only accepted by unreliable buyers under cash in advance.* Then, the seller offers $P^{CIA,S3} = \frac{\tilde{\eta}R}{(1+r_d)}$, which is only accepted by unreliable buyers. The seller expected profits is:

$$E[\Pi_{RS}^{CIA,S3}] = (1 - \eta^*)(1 + r_d) \left(\frac{\tilde{\eta}R}{(1 + r_d)} - C \right)$$

The seller prefers the pooling equilibrium where she serves both reliable and unreliable buyers as long as:

$$R \tilde{\eta}(\tilde{\eta}^* - (1 - \eta^*)) \geq C(1 + r_d)\eta^* \quad (\text{C.6})$$

Note that a necessary condition to ensure a pooling equilibrium is that $\tilde{\eta}^* > (1 - \eta^*)$. This can be rewritten to:

$$\eta^* > \frac{1 - \phi}{2 - \phi}$$

Then, (C.6) can be rewritten as:

$$\frac{R}{C} \geq \frac{(1 + r_d)\eta^*}{\tilde{\eta}(\tilde{\eta}^* - (1 - \eta^*))} \quad (\text{C.7})$$

Cash in Advance, Separating Case 4 *The reliable seller chooses trade credit, but the unreliable seller chooses cash in advance.* Then, the buyer knows that she is dealing with an unreliable seller and the participation constraint becomes:

$$\mathbb{E}[\Pi_{RB}^{CIA, S4}] = \phi R - (1 + r_b^*(\tilde{\eta}^*))P^{CIA, S4}.$$

The unreliable seller then picks the optimal payment $P^{CIA, S4} = \frac{\phi}{1 + r_b^*(\tilde{\eta}^*)} R$, delivering expected profits of:

$$\mathbb{E}[\Pi_{US}^{CIA, S4}] = (1 + r_d)\phi \left(\frac{R}{1 + r_b^*(\tilde{\eta}^*)} - C \right).$$

A sufficient condition for the pooling case to dominate is:

$$\Pi_{US}^{TC,P} - \Pi_{US}^{CIA,SC4} \geq \Pi_{RS}^{TC,P} - \Pi_{RS}^{CIA,P}$$

Which can be rewritten to:

$$\Pi_{US}^{TC,P} - \Pi_{RS}^{TC,P} \geq \Pi_{US}^{CIA,SC4} - \Pi_{RS}^{CIA,P}$$

Plugging in the profits and simplifying delivers:

$$\frac{R}{C} > \frac{(1 + r_b^*(\tilde{\eta}^*))}{\eta(1 - \phi)} \left[(1 - \phi) - \frac{(1 - \phi)}{\tilde{\eta}} \right]$$

Can cancel further to get:

$$\frac{R}{C} \geq \frac{(1 + r_b^*(\tilde{\eta}^*))}{\eta} \left[1 - \frac{1}{\tilde{\eta}} \right]$$

This condition holds if:

$$\tilde{\eta} \leq 1 \tag{C.8}$$

Combining the conditions To summarize, pooling requires the following three conditions:

$$\frac{R}{C} \geq \frac{\eta^*(1 + r_d)}{\tilde{\eta}(\eta^* - (1 - \eta^*)(1 - \phi))} \tag{C.9}$$

$$\eta \geq \frac{1 - \phi}{2 - \phi} \tag{C.10}$$

$$\frac{R}{C} \geq \frac{\eta^*(1 + r_d)}{\tilde{\eta}(\eta^* - (1 - \eta^*)(1 - \phi))} \tag{C.11}$$

$$\tilde{\eta} \leq 1. \tag{C.12}$$

Now, rewriting equation (C.9), focusing on the symmetric case:

$$\frac{\eta}{\tilde{\eta}(\tilde{\eta} - (1 - \eta))} \leq \tilde{X}$$

And taking the derivative with respect to η delivers:

$$\frac{\tilde{\eta}(\tilde{\eta} - (1 - \eta)) - \eta((1 - \phi)(\tilde{\eta} - (1 - \eta)) + \tilde{\eta}(2 - \phi))}{(\tilde{\eta}(\tilde{\eta} - (1 - \eta)))^2} < 0$$

This can be simplified to:

$$-(1 - \phi)[\eta^2(2 - \phi) + \phi] < 0 \quad (\text{C.13})$$

Thus, we now know that condition (C.9) gets weaker as η increases. So there is always a level of η for which the condition holds.

Finally, for $\eta \rightarrow 1$, the above conditions converge to:

$$\frac{R}{C} > 1 + r_d \quad (\text{C.14})$$

$$1 > 0 \quad (\text{C.15})$$

$$\frac{R}{C} > 1 + r_d \quad (\text{C.16})$$

We thus know, that there exists an $\eta, \eta^* > 0$ for which all pooling conditions hold. Intuitively, as the fraction of unreliable firms converges to zero, it is always optimal to offer contracts that are also acceptable to reliable firms to maximize expected profits.

D Model Extensions

D.1 Trade Credit Maturity

Proof for proposition 3

Part (i): At $t=0$, the FOC, given by equation (14), is strictly positive, as only the financing cost channel is active:

$$\frac{\partial E[\Pi_S^{TC,I}]/C}{\partial t}|_{t=0} = (r_b - r_d) \left(\frac{\mu}{(1 + r_b T)^2} - 1 \right) > 0.$$

At $t=T$, equation (14) simplifies to:

$$\frac{\partial E[\Pi_S^{TC,I}]/C}{\partial t}|_{t=T} = (r_b - r_d) (\tilde{\eta}\mu - 1) + \mu\tilde{\eta}(T)'$$

This expression is negative if:

$$\tilde{\eta}(T)' < -\frac{r_b - r_d}{\mu} (\tilde{\eta}\mu - 1)$$

Due to continuity, this implies that the first order condition is zero at least once between $t = 0$ and $t = T$. To ascertain that we have a unique solution that represents a maximum, we look at the second-order condition of the problem next. Taking derivatives of equation (14) with respect to t we find:

$$\begin{aligned} \frac{\partial^2(\Pi^{TC,I}/C)}{\partial t^2} &= (r_b - r_d) \left[\frac{\tilde{\eta}(t)'\mu}{(1 + r_b(T-t))^2} + \frac{2\tilde{\eta}(t)\mu r_b}{(1 + r_b(T-t))^3} \right] \\ &+ \tilde{\eta}(t)''\mu \frac{(1 + r_d(T-t))}{(1 + r_b(T-t))} - \frac{\tilde{\eta}(t)'\mu r_d}{(1 + r_b(T-t))} \\ &+ \tilde{\eta}(t)'\mu r_b \frac{(1 + r_d(T-t))}{(1 + r_b(T-t))^2} \end{aligned} \tag{D.1}$$

Gathering terms:

$$\frac{\partial^2(\Pi^{TC,I}/C)}{\partial t^2} = \frac{2(r_b - r_d)\mu}{(1 + r_b(T - t))^2} \left[\tilde{\eta}(t)' + \frac{\tilde{\eta}(t)r_b}{(1 + r_b(T - t))} \right] + \underbrace{\tilde{\eta}(t)''\mu \frac{(1 + r_d(T - t))}{(1 + r_b(T - t))}}_{<0 \text{ as } \tilde{\eta}(t)'' < 0}$$

A sufficient condition for $\frac{\partial^2(\Pi^{TC,I}/C)}{\partial t^2} < 0$ is that the term in the square brackets is negative:

$$\tilde{\eta}(t)' + \frac{\tilde{\eta}(t)r_b}{(1 + r_b(T - t))} < 0 \quad (\text{D.2})$$

This can be rewritten to:

$$-\frac{\tilde{\eta}(t)'}{\tilde{\eta}(t)} > \frac{r_b}{1 + r_b(T - t)}, \quad (\text{D.3})$$

which is one of the conditions we required for the function $\tilde{\eta}(t)$. Intuitively, this condition requires that the diversion risk rises sufficiently quickly with maturity, t . When this condition holds, the SOC is always negative in the range $t \in [0, T]$ and there exists a unique interior solution for t that maximizes expected seller profits.

Part (ii): The cross-derivative of expected seller profits over C w.r.t. t and μ is given by:

$$\frac{\partial^2 \Pi^{TC,I}}{\partial t \partial \mu} / C = (r_b - r_d) \left(\frac{\tilde{\eta}(t)}{(1 + r_b(T - t))^2} \right) + \frac{1}{1 + r_b(T - t)} [(1 + r_d(T - t))(\tilde{\eta}(t)')]$$

As long as the financing cost advantage effect weakly dominates the diversion effect, it is the case that:

$$\frac{1}{1 + r_b(T - t)} [(1 + r_d(T - t))(\tilde{\eta}(t)')] \geq -\frac{r_b - r_d}{\mu} \left(\frac{\tilde{\eta}(t)\mu}{(1 + r_b(T - t))^2} - 1 \right).$$

Substitute this expression in to get:

$$\begin{aligned}
\frac{\partial^2 \Pi^{TC,I}}{\partial t \partial \mu} / C &= (r_b - r_d) \left(\frac{\tilde{\eta}(t)}{(1 + r_b(T - t))^2} \right) + \frac{1}{1 + r_b(T - t)} [(1 + r_d(T - t))(\tilde{\eta}(t)')] \\
&\geq (r_b - r_d) \left(\frac{\tilde{\eta}(t)}{(1 + r_b(T - t))^2} \right) - \frac{r_b - r_d}{\mu} \left(\frac{\tilde{\eta}(t)\mu}{(1 + r_b(T - t))^2} - 1 \right) \\
&= \frac{r_b - r_d}{\mu} > 0.
\end{aligned}$$

That is, the effect of the maturity on profits increases in the markup μ . In addition, this expression increases in the borrowing rate r_b , implying that the effect of the markup on the optimal maturity increases in r_b .

D.2 Variable Markups

This subsection presents additional details and derivations for the variable markup extension presented in section 2.4.1. Let the linear demand take the form $Q(p) = 1 - p$. Profits can be represented by: $\Pi = \alpha p Q(p) - \beta c Q(p) = (\alpha p - \beta c)(1 - p)$. With: $\alpha^{TC} = 1$; $\beta^{TC} = 1 + r_b$; $\alpha^{CIA} = \frac{1+r_d}{1+r_b^*}$; $\beta^{CIA} = 1 + r_d$. Solving for the optimal price charged to final consumers, we find: $p = \frac{1}{2} + \frac{\beta c}{2\alpha}$; $p^{TC} = \frac{1}{2} + \frac{(1+r_b)c}{2}$; $p^{CIA} = \frac{1}{2} + \frac{(1+r_b^*)c}{2}$. Then, calculate firm-to-firm markups, recalling that $P^{TC} = R$ and $P^{CIA} = \frac{R}{1+r_b^*}$ as $\mu^{TC} = \frac{R}{Q_c} = \frac{p}{c}$; $\mu^{CIA} = \frac{1}{1+r_b^*} \frac{R}{Q_c} = \frac{1}{1+r_b^*} \frac{p}{c}$.¹

Which delivers:

$$\mu^{TC} = \frac{1}{2c} + \frac{1 + r_b}{2}, \quad (D.4)$$

$$\mu^{CIA} = \frac{1}{1 + r_b^*} \left(\frac{1}{2c} + \frac{1 + r_b^*}{2} \right) \quad (D.5)$$

¹Note that the markup of interest is the one charged to the buying firm (P/Q_c) as opposed to the markup charged to the final consumer p/c .

It is easy to see that markups decrease (increase) in the marginal cost (productivity). We can now derive profits as $\Pi = \alpha \left(\frac{1}{2} - \frac{\beta c}{2\alpha} \right)^2$; $\Pi^{TC} = \left(\frac{1}{2} - \frac{(1+r_b)c}{2} \right)^2$; $\Pi^{CIA} = \frac{1+r_d}{1+r_b^*} \left(\frac{1}{2} - \frac{(1+r_b^*)c}{2} \right)^2$. From this we can calculate the difference in profits between trade credit and cash in advance as:

$$\begin{aligned} \Delta\Pi &= \left[\left(\frac{1}{2} - \frac{(1+r_b)c}{2} \right)^2 - \frac{1+r_d}{1+r_b^*} \left(\frac{1}{2} - \frac{(1+r_b^*)c}{2} \right)^2 \right] \\ &= \left[\left(\mu^{TC}(c) - (1+r_b) \right)^2 - (1+r_d)(1+r_b^*) \left(\mu^{CIA}(c) - 1 \right)^2 \right] c^2 \end{aligned} \quad (D.6)$$

Taking the derivative with respect to c delivers $\frac{\partial \Delta\Pi}{\partial c} = -(1+r_b) \left(\frac{1}{2} - \frac{(1+r_b)c}{2} \right) + (1+r_d) \left(\frac{1}{2} - \frac{(1+r_b^*)c}{2} \right)$. It is easy to see that the derivative is negative as long as $r_b > r_d$ and r_b^* and r_b are not too different. To see this, redefine $\left(\frac{1}{2} - \frac{(1+r_b^*)c}{2} \right) = \left(\frac{1}{2} - \frac{(1+r_b)c}{2} \right) + \epsilon$, with: $\epsilon = \frac{c}{2}(r_b - r_b^*)$. Then the condition simplifies to:

$$\begin{aligned} \frac{\partial \Delta\Pi}{\partial c} &= -(r_b - r_d) \left(\frac{1}{2} - \frac{(1+r_b)c}{2} \right) + (1+r_d)\epsilon \\ &= -(r_b - r_d) \left(\mu^{TC}(c) - (1+r_b) \right) c + (1+r_d)\epsilon \end{aligned} \quad (D.7)$$

So as long as $r_b > r_d$, $\mu^{TC} > 1 + r_b$, and ϵ is not too large, this derivative will be negative. As shown above, ϵ is a function of the difference in borrowing rates abroad and at home and goes to zero in the case of symmetric borrowing costs.

D.3 Partial Pre-Payments

This section provides the details on the partial pre-payments extension and shows that the only partial payment that can be optimal is one that equals the production costs, C . There are two cases to consider.

Case 1 In the first case, the buyer pays at least the production cost C in advance ($\chi P^{PP} \geq C$, where $\chi \leq 1$ denotes the fraction of). Then, profits can be written as $\Pi_S^{PP} = (1 + r_d)(\chi P^{PP} - C) + (1 - \chi)P^{PP}$; $\Pi_B^{PP} = R - (1 + r_b^*)\chi P^{PP} - (1 - \chi)P^{PP}$. Solving for the maximum payment that satisfies the participation constraint of the buyer implies $P^{PP} = \frac{R}{1 + \chi r_b^*}$. Plugging P^{PP} back into seller profits gives $\Pi_S^{PP} = (1 + r_d) \left(\frac{\chi R}{1 + \chi r_b^*} - C \right) + \frac{(1 - \chi)R}{1 + \chi r_b^*}$. Then, taking the derivative with respect to χ delivers:

$$\frac{\partial \Pi_S^{PP}}{\partial \chi} = -(r_b^* - r_d) \frac{R}{(1 + \chi r_b^*)^2} \quad (\text{D.8})$$

Equation (D.8) implies that profits fall in the pre-payment share if the foreign borrowing rate exceeds the deposit rate, $r_b^* > r_d$. Thus, if $r_b^* > r_d$, the optimal pre-payment is less or equal to production costs, C . If borrowing abroad is very cheap and $r_b^* < r_d$, equation (D.8) becomes positive and full pre-payment (cash in advance) is optimal.

Case 2 In the second case, the buyer pays less than C in advance ($\chi P^{PP} < C$). The problem then reads $\Pi_S^{PP} = (1 + r_b)(\chi P^{PP} - C) + (1 - \chi)P^{PP}$; $\Pi_B^{PP} = R - (1 + r_b^*)\chi P^{PP} - (1 - \chi)P^{PP}$. As the buyer profits do not change, the payment remains $P = \frac{R}{1 + \chi r_b^*}$. Plugging into seller profits delivers $\Pi_S^{PP} = (1 + r_b) \left(\frac{\chi R}{1 + \chi r_b^*} - C \right) + \frac{(1 - \chi)R}{1 + \chi r_b^*}$. Taking the derivative with respect to χ delivers:

$$\frac{\partial \Pi_S^{PP}}{\partial \chi} = (r_b - r_b^*) \frac{R}{(1 + \chi r_b^*)^2} \quad (\text{D.9})$$

Equation (D.9) is driven by the difference in borrowing rates, $r_b - r_b^*$. If the domestic borrowing rate exceeds the foreign borrowing rate, the optimal pre-payment is greater or equal to the production costs. If the foreign borrowing rate is higher than the domestic borrowing rate, the optimal pre-payment is zero.

The optimal pre-payment To summarize, there are three cases:

- i) Suppose $r_b^* > r_b$. Then, $r_b^* > r_d$ because $r_b > r_d$. And equation (D.8) implies $\chi^{PPP} \leq C$ and equation (D.9) implies $\chi^{PPP} = 0 \Rightarrow \chi^{PPP} = 0$. The seller provides trade credit.
- ii) Suppose $r_b^* > r_d$ and $r_b > r_b^*$. Then, equation (D.8) implies $\chi^{PPP} \leq C$ and equation (D.9) implies $\chi^{PPP} \geq C \Rightarrow \chi^{PPP} = C$. The seller asks for a pre-payment of C .
- iii) Suppose $r_d > r_b^*$. Then, $r_b > r_b^*$ because $r_b > r_d$. And equation (D.8) implies that $\chi = 1$ and equation (D.9) implies that $\chi^{PPP} \geq C \Rightarrow \chi^{PPP} = 1$. The seller asks for cash in advance.

D.4 Nash Bargaining

This section provides details for the Nash-Bargaining extension. The bargaining model is solved in two steps. First, profits under the two payment options are derived. Then, firms pick the payment option that maximizes joint surplus. Let θ ($1 - \theta$) be the bargaining power of the seller (buyer).

Trade Credit With trade credit, the bargaining problem reads:

$$NP^{TC} = (\Pi_S^{TC})^\theta (\Pi_B^{TC})^{1-\theta} = \underbrace{(P^{TC} - (1 + r_b)C)}_{\text{Seller Profit}}^\theta \underbrace{(R - P^{TC})}_{\text{Buyer Profit}}^{1-\theta}.$$

Solving the problem delivers an optimal payment $P^{TC} = \theta R + (1 - \theta)(1 + r_b)C$ and a Nash Product with trade credit of:

$$NP^{TC} = \theta^\theta (1 - \theta)^{1-\theta} (R - (1 + r_b)C). \quad (\text{D.10})$$

Cash in Advance With cash in advance, the bargaining problem reads:

$$NP^{CIA} = (\Pi_S^{CIA})^\theta (\Pi_B^{CIA})^{1-\theta} = \underbrace{[(1+r_d)(P^{CIA} - C)]^\theta}_{\text{Seller Profit}} \underbrace{(R - (1+r_b^*)P^{CIA})^{1-\theta}}_{\text{Buyer Profit}}.$$

Solving the problem delivers an optimal payment $P^{CIA} = \frac{\theta R + (1-\theta)(1+r_b^*)C}{1+r_b^*}$ and Nash product under cash in advance:

$$NP^{CIA} = \theta^\theta (1-\theta)^{1-\theta} (1+r_d)^\theta (1+r_b^*)^{-\theta} (R - (1+r_b^*)C) \quad (\text{D.11})$$

Combining equations (D.10) and (D.11), the two firms prefer trade credit if:

$$(\mu - (1+r_b))(1+r_b^*)^\theta - (\mu - (1+r_b^*))(1+r_d)^\theta > 0 \quad (\text{D.12})$$

Proof for predictions from Proposition 1 Suppose the foreign borrowing rate is above the domestic deposit rate ($r_b^* > r_d$). and the seller charges a positive markup over effective costs ($\mu > 1+r_b$). Then:

- i) If the buyer and seller face equal borrowing costs ($r_b = r_b^*$), the seller always prefer trade credit.
- ii) There is always a markup, μ , that is large enough to make the seller choose trade credit over cash in advance.

Proof. i) if $r_b = r_b^*$, then condition (D.12) simplifies to: $(\mu - (1+r_b))((1+r_b)^\theta - (1+r_d)^\theta) > 0$.

Under the assumption stated in the proposition, trade credit is then always preferred over cash in advance, as long as $\theta > 0$.

ii) let μ go to infinity. Then, condition (D.12) becomes: $(1+r_b^*)^\theta - (1+r_d)^\theta > 0$, which always holds, as long as $\theta > 0$. ■

Proof for predictions from Proposition 2 Suppose $r_b^* > r_d$. Then:

- i) The use of trade credit increases with the markup μ .
- ii) This effect increases with r_b^* and decreases with r_d .

Proof. i) Taking the derivative of condition (D.12) with respect to μ delivers: $\frac{\partial Equ.(D.12)}{\partial \mu} = (1 + r_b^*)^\theta - (1 + r_d)^\theta$. This derivative is positive if $r_b^* > r_d$ and $\theta > 0$. ii) Taking the cross derivatives with respect to μ and $1 + r_b^*$ and $1 + r_d$, respectively, delivers: $\frac{\partial Equ.(D.12)^2}{\partial \mu \partial (1 + r_b^*)} = \theta(1 + r_b^*)^{\theta-1}$ and $\frac{\partial Equ.(D.12)^2}{\partial \mu \partial (1 + r_d)} = -\theta(1 + r_d)^{\theta-1}$. These two cross-derivatives are positive and negative, respectively, as long as θ is larger than zero. ■

D.5 Letters of Credit

This section provides details for the extension with letters of credit that builds on the model with diversion risk in section 2.2.

Letter of Credit Letters of credit are a payment form that is used exclusively in international trade transactions. With a letter of credit, banks serve as intermediaries in the transaction to resolve diversion problems between buyers and sellers. Assume that a bank can incur monitoring costs to perfectly verify delivery of goods before paying out funds to the seller. For this service, the buyer pays a fee to the bank and commits to paying the seller.² Assume that this fee is proportional to the transaction size: $F^{LC} = f^{LC} P^{LC}$. The seller only receives payment from the bank after providing proof of shipment or delivery. Assuming that firms are still able to divert bank funds as before, profits are given by: $\Pi_S^{LC} = P^{LC} - (1 + r_b(\tilde{\eta}))C$ and $\Pi_B^{LC} = R - P^{LC} - (1 + r_b^*(\tilde{\eta}^*))(f^{LC} P^{LC})$. With a letter of credit, there is no risk and the seller

²This commitment can either reflect a long-term relationship with the bank or may require a deposit in the bank up to the value of the letter of credit. For tractability, we assume that it is sufficient for the buyer to pay the letter of credit fee in advance.

receives P^{LC} with certainty and the buyer generates revenues R with certainty.³ Solving for the optimal P^{LC} that makes the buyer indifferent delivers $P^{LC} = \frac{R}{1+f^{LC}(1+r_b^*(\tilde{\eta}^*))}$ And plugging back into seller profits leads to:

$$\Pi_S^{LC} = \frac{R}{1+f^{LC}(1+r_b^*(\tilde{\eta}^*))} - (1+r_b(\tilde{\eta}))C.$$

Optimal Payment Choice Comparing trade credit with a letter of credit delivers:

$$E[\Pi_S^{TC}] - E[\Pi_S^{LC}] = \left[\tilde{\eta}^* - \frac{1}{1+f^{LC}(1+r_b^*(\tilde{\eta}^*))} \right] \mu C > 0, \quad (\text{D.13})$$

As stated in proposition 6, the markup, μ , does not affect the sign of this equation and therefore has no effect on the choice between trade credit and a letter of credit.

³This is a simplifying assumption, as, in reality, letters of credit are not completely risk-free. Relaxing this assumption should not affect any of our results. For a detailed analysis of letter of credit risk see [Niepmann and Schmidt-Eisenlohr \(2017b\)](#).

E Additional Details on Markups Estimation

To test the predictions of the theory, we compute markups at the seller-product level using the methodology proposed by [De Loecker et al. \(2016\)](#). The main advantage of this methodology is that it allows us to compute markups abstracting from market-level demand information. It only requires to assume that firms minimize cost for each product and that at least one input is fully flexible.

The starting point in [De Loecker et al. \(2016\)](#), is to consider the firm's cost minimization problem. After rearranging the first-order condition of the problem for any flexible input V , the markup of product p produced by firm i in year t (μ_{ipt}) can be computed as the ratio between the output elasticity of product p with respect to the flexible input V (θ_{ipt}^V) and expenditure share of the flexible input V (relative to the sales of product p ; $s_{ipt}^V \equiv P_{ipt}^V V_{ipt} / P_{ipt} Q_{ipt}$):

$$\underbrace{\mu_{ipt}}_{\text{Markup}} \equiv \frac{P_{ipt}}{MC_{ipt}} = \frac{\theta_{ipt}^V}{s_{ipt}^V}, \quad (\text{E.1})$$

where P (P^V) denotes the price of output Q (input V), and MC is marginal cost. While the numerator of equation (E.1) – the input-output elasticity of product p – needs to be estimated, the denominator is directly observable in our data. Next, we explain the procedure we follow for deriving each of these elements.

Input-output elasticity. To estimate the input-output elasticities, we specify production functions for each product p using labor (L), capital (K), and materials (M) as production inputs:

$$Q_{ipt} = \Omega_{ipt} F(K_{ipt}, L_{ipt}, M_{ipt}) \quad (\text{E.2})$$

where Q is physical output, and Ω denotes productivity. There are two important assumptions on equation (E.2). First, the production function is product-specific, which implies that single and multi-product firms use the same technology to produce a given product. Second, as is standard in the estimation of production functions, we assume Hicks-Neutrality, so that Ω is log-additive.

The estimation of (E.2) follows De Loecker et al. (2016) in using the subset of single-product firms to identify the coefficients of the production function. The reason for using only single-product firms is that, for this set of firms, there is no need of specifying how inputs are distributed across individual outputs. Different from De Loecker et al., we deflate inputs expenditure with firm-specific input price indexes to avoid that the so-called input price bias affect the estimated coefficients (see De Loecker and Goldberg, 2014).⁴

Our baseline specification assumes a Cobb-Douglas production function, and allows for the presence of a log-additive non-anticipated shock (ε). A shortcoming of the Cobb-Douglas specification is that it assumes that input-output elasticities are constant across firms and over time. On the other hand, the Cobb-Douglas specification is widely used, allowing for a more direct comparison of our results with other estimates in the literature. In the robustness checks section, we present results derived with a more flexible Translog production function, which allows for different types of complementarities among production inputs. Results are quantitatively similar, although coefficients are slightly less precisely estimated than with the Cobb-Douglas baseline. Taking logs to (E.2), we obtain (lower cases denote logarithm of the variables)

$$q_{ipt} = \alpha_k^p k_{ipt} + \alpha_l^p l_{ipt} + \alpha_m^p m_{ipt} + \omega_{ipt} + \varepsilon_{ipt} \quad (\text{E.3})$$

⁴In De Loecker et al. (2016), input prices are not available in their sample of Indian firms, so they implement a correction to control for input price variation. We discuss below the construction of the input price index we use in our sample of Chilean firms.

The estimation of (E.3) follows [Akerberg et al. \(2015\)](#) (henceforth, ACF), who extend the methodology proposed by [Olley and Pakes \(1996\)](#) and [Levinsohn and Petrin \(2003\)](#) to control for the endogeneity of firms' inputs choice –which is based on the actual level of firms' productivity.⁵ To identify the coefficients of the production function, we build moments based on the productivity innovation ξ . We specify the following process for the law of motion of productivity:

$$\omega_{ipt} = g(\omega_{ipt-1}, d_{ipt-1}^x, d_{ipt-1}^i, d_{ipt-1}^x \times d_{ipt-1}^i, \hat{s}_{ipt-1}) + \xi_{ipt} \quad (\text{E.4})$$

where d^x is an export dummy, d^i is a categorical variable for periods with positive investment, and $\hat{s}$ is the probability that the firm remains single-product. The endogenous productivity process (E.4) allows the firms' productivity path to be affected by past exporting and investment decisions. In addition, it follows [De Loecker et al. \(2016\)](#) in including the probability of remaining single-product to correct for the bias that results from firm switching non-randomly from single to multi-product.

The first step of the ACF procedure involves expressing productivity in terms of observables. To do so, we use inverse material demand $h_t(\cdot)$ as in [Levinsohn and Petrin \(2003\)](#) to proxy for unobserved productivity, and estimate expected output $\phi_t(k_{ipt}, l_{ipt}, m_{ipt}; \mathbf{x}_{ipt})$ to remove the unanticipated shock component ε_{ipt} from (E.3).⁶ Then, the ACF procedure exploits this representation to express productivity as a function of data and parameters: $\omega_{ipt}(\boldsymbol{\alpha}) = \hat{\phi}_t(\cdot) - \alpha_k k_{ipt} - \alpha_l l_{ipt} - \alpha_m m_{ipt}$, and form the productivity innovation ξ_{ipt} from (E.4) as a function of the parameters $\boldsymbol{\alpha}$. The second step of ACF routine forms moment conditions on ξ_{ipt} to identify

⁵ACF show that the labor elasticity is in most cases unidentified by the two-stage method of [Olley and Pakes \(1996\)](#) and [Levinsohn and Petrin \(2003\)](#).

⁶The vector $\mathbf{x}_{ipt}$ includes other variables affecting material demand, such as time and product dummies. We approximate $\phi_t(\cdot)$ with a full second-degree polynomial in capital, labor, and materials.

all parameters α through GMM:

$$\mathbb{E}(\xi_{ipt}(\alpha) \cdot \mathbf{Z}_{ipt}) = 0 \quad (\text{E.5})$$

where $\mathbf{Z}_{ipt}$ contains lagged materials, labor, and capital, and current capital. Once the parameters are estimated, the input-output elasticities are recovered for each product as $\theta_{ipt}^V \equiv \partial \ln Q_{ipt} / \partial \ln V_{ipt}$. For the Cobb-Douglas case, $\theta_{ipt}^V = \alpha_V^p$, so that the input-output elasticity is constant for all plants producing a given product p .⁷

Implementation. To derive markups, we use materials as the relevant flexible input to compute the output elasticity. While in principle, labor could also be used to compute markups, the existence of long-term contracts and firing costs make firms less likely to adjust labor after the occurrence of shocks. The second component needed in (E.1) to compute markups is the expenditure share, which requires to identify the assignment of firms' inputs across outputs produced by the firm. To implement this, we follow [Garcia-Marin and Voigtländer \(2019\)](#) and exploit a unique feature of our data: ENIA provides information on total variable costs (labor cost and materials) for each product produced by the firms. We use this information to proxy for product-specific input use assuming that inputs are used approximately in proportion to the variable cost shares, so that the value of materials' expenditure $M_{ipt} = P_{ipt}^V V_{ipt}$ is computed as

$$\tilde{M}_{ipt} = \rho_{ipt} \cdot \tilde{M}_{it}, \quad \text{where} \quad \rho_{ipt} = \frac{TV C_{ipt}}{\sum_j TV C_{ijt}}. \quad (\text{E.6})$$

⁷In the Translog case, the input elasticities θ_{ipt}^V depend on the firms' input use. For multi-product firms, we derive inputs' use by each output following the same procedure we apply for computing the expenditure share of the inputs s_{ipt}^V explained next.

Finally, we compute the expenditure share by dividing the value of material inputs by product-specific revenues, which are observed in the data.

Input Price Index. To avoid input price bias in the estimation of the production function parameters (see [De Loecker and Goldberg, 2014](#), for details), we deflate materials’ expenditure using firm-specific price indexes. The construction of the input price deflator involves five steps. First, we define the unit value of input p purchased by firm i in period t as $P_{ipt} = V_{ipt}/Q_{ipt}$, where V_{ipt} denotes input p value, and Q_{ipt} denotes the corresponding quantity purchased. Next, we calculate the (weighted) average unit value of input p across all firms purchasing the input in year t . Then, for each firm, we compute the (log) price deviation from the (weighted) average for all the inputs purchased by the firm in year t . The next step involves averaging the resulting price deviations at the firm level, using inputs’ expenditure as weight. Finally, we anchor the resulting average firm-level input price deviation to aggregate (4-digit) input price deflators provided by the Chilean statistical agency. Therefore, the resulting input price index reflects both, changes in the aggregate input price inflation, as well as firm-level heterogeneity in the price paid by firms for their inputs.

F Data Appendix

In this appendix we provide additional details on the construction of the dataset we use in the main empirical analysis. In the following, we briefly discuss the procedure we follow to combine the production data in ENIA with the customs-level data at the firm-product level. We also explain the data cleaning procedure we apply to avoid inconsistencies.

The main issue in combining data from Customs and ENIA at the firm-product level is that products are classified using different nomenclatures in both datasets: ENIA classifies products according to the Central Product Classification (CPC), while the Chilean Customs

Administration classifies products according to the Harmonized System (HS). To deal with this issue, we follow several steps. First, we use the United Nations' correspondence tables to determine the list of HS products that could potentially be matched to each CPC product in ENIA.⁸ We then merge the resulting dataset with customs data at the firm-HS-year level. This procedure results in two cases: (i) All exported HS products in customs within a firm-year pair are merged to ENIA, and (ii) Only a fraction (or none) of the exported products are matched to ENIA within a firm-year pair. For the latter cases, whenever there is concordance within 4-digit HS categories, we manually merge observations based on HS and CPC product descriptions. Borderline cases (no clear connection between product descriptions), as well as cases with no concordance at the 4-digit HS level are dropped.

In addition, to ensure a consistent dataset, we follow several steps. In particular, we exclude: (i) firm-year observations that have zero values for raw materials expenditure or employment, (ii) firm-product-year observations with zero or missing sales, product quantities, or with extreme values for markups (above the 98th or below the 2nd percentiles, or with large implausible variations in markups within firm-products), and (iii) destination-year pairs with extreme values of the real ex-post borrowing rates, to avoid the influence of extreme values resulting from inflationary or deflationary episodes.⁹ The final dataset consists of 88,546 firm-product-destinations-year observations. The sample represents 69.2% of the value of Chilean (non-copper) exports over the period 2003-2007. Table F.6 presents the estimated markups at the level of 2-digit industries.

G Additional Details on Robustness Checks

In this section, we provide details on the robustness checks mentioned in section 5.2:

⁸The correspondence table establishes matches between 5-digit CPC and 6-digit HS products. This level of disaggregation corresponds to 783 5-digit CPC products.

⁹In practice, this correction drops country-years with real borrowing rates above 35%, and below -4%.

Table F.6. Estimated Markups

Product	Mean	Median	St. Deviation
Food and Beverages	1.267	1.132	0.509
Textiles	1.543	1.432	0.562
Apparel	1.278	1.254	0.469
Wood and Furniture	1.127	1.009	0.435
Paper	1.157	1.042	0.462
Basic Chemicals	1.364	1.162	0.685
Plastic and Rubber	1.218	1.080	0.505
Non-Metallic Manufactures	1.657	1.541	0.785
Metallic Manufactures	1.166	0.995	0.505
Machinery and Equipment	1.131	0.989	0.480
Total	1.255	1.110	0.538

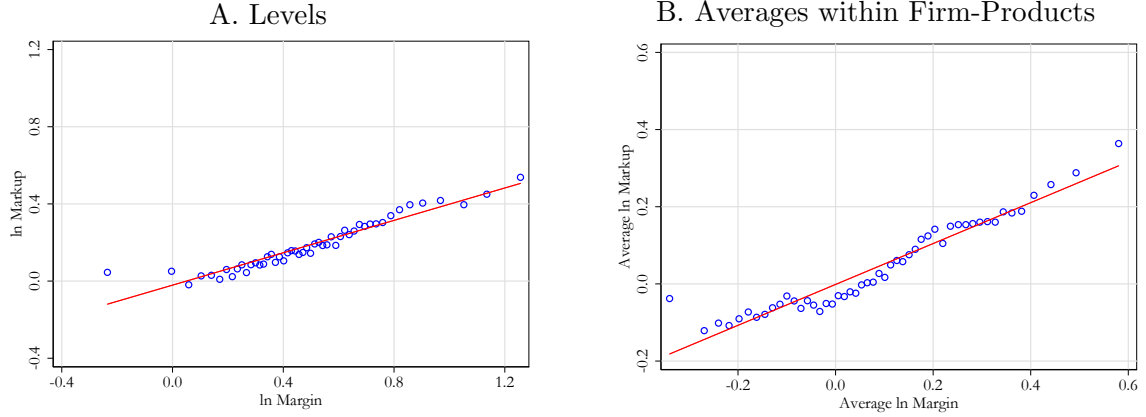
Notes: This table reports the average markup by aggregate sector for the sample Chilean exporters over the period 2003-2007.

Average product margin. An additional proxy for markups that we can compute in our sample is product-level price-cost margins. ENIA reports the variable production cost per product, defined as the sum of raw material and direct labor costs involved in the production of each product. Product margins can be derived by dividing prices (unit values) over this reported measure of average variable cost. Note that the average variable cost is self-reported by managers, making the application of rules of thumb likely.

Figure G.1 shows binscatter plots for firm-product markups and sales-cost margins (with products defined at the HS-8 level), for the raw data (left panel), and averaging across observations within firm-product pairs (right panel). Both figures control for country-year fixed effects (that is, the figure plots the within plant-product variation that we exploit empirically). There is a remarkable positive relationship between markups and reported margins, suggesting that our markup estimates yield sensible information about the profitability of the products produced by the firm. This lends strong support to the markup-based methodology for backing out marginal costs by [De Loecker et al. \(2016\)](#). In addition, there seems to be a tighter relationship between markups and margins when both variables are averaged within firm-products.¹⁰

¹⁰One reason why both measures could be more correlated over longer periods of time is that the sales-cost margin measure relies on self-reported average variable cost. If managers measure product-level variable costs

Figure G.1. Firm-Product level Markup and Sales-Cost Margin



Notes: The figure plots a binscatters diagram for firm-product markups and sales-cost margins.

First Stage Estimates Table G.1 shows first stage estimates for the IV specifications in table 16 (columns 4-6), where we instrument firm-product markups and its interaction with the borrowing rate with firm-product physical productivity (TFPQ) and its interaction with the borrowing rate. Across specifications, we obtain strong first stages, with stable coefficients for the first stage regression for the interaction between markups and the borrowing rate.

Accounting for the Domestic Deposit Rate. Table G.2 replicates table 3 when interacting the markup with the difference between the foreign borrowing rate and the domestic deposit rate. Estimated coefficients are very similar to the ones presented in table 3. In fact, in columns 2, 4, 6, and 8, the firm-product-year FEs eliminate any effect of rd and its interaction with the markup, and results are therefore identical for those columns.

with error, then the sales-cost margin may be a poorer approximation of markups in the short run. However, if managers do not make systematic mistakes when reporting average variable costs, the measurement error cancels out when averaging over longer periods.

Table G.1. First Stage Regressions, Table 3

Specification	Spec. (3)		Spec. (4)	Spec. (7)		Spec. (8)
	(1)	(2)	(3)	(4)	(5)	(6)
	$\ln(\text{markup})$	$\ln(\text{markup}) \times r_b^*$	$\ln(\text{markup}) \times r_b^*$	$\ln(\text{markup})$	$\ln(\text{markup}) \times r_b^*$	$\ln(\text{markup}) \times r_b^*$
$\ln(\text{TFPQ})$	0.0522*** (0.0028)	0.1166*** (0.0290)	—	0.0558*** (0.0033)	0.1326*** (0.0318)	—
$\ln(\text{TFPQ}) \times r_b^*$	-0.0002*** (0.0001)	0.0298*** (0.0038)	0.0313*** (0.0045)	-0.0002** (0.0001)	0.0306*** (0.0042)	0.0322*** (0.0048)
First Stage F-statistic	172.5		49.5	21.1		44.7
Firm-Year FE	Yes	Yes	—	Yes	Yes	—
Product FE	Yes	Yes	—	Yes	Yes	—
Country-Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm-Product-Year FE	—	—	Yes	—	—	Yes
Observations	88,546	88,546	88,546	77,328	77,328	77,328

Notes: The table show first-stage regressions for the IV specifications (equation 16 in the main text) in table 3. Columns 1 and 2 show the first stage regressions for the two instrumented variables used in specification (3). Column 3 shows the first stage regression for the interaction term between markups (in logs) and the borrowing rate – the only variable not absorbed by the fixed effects – in specification (4). Columns 4 and 5 show the first stage regressions for specification (7). Finally, column 6 shows the first stage regression for specification (8). All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Markups and TFPQ are computed at the firm-product level (products are defined at the 5-digit CPC level). The (cluster-robust) Kleibergen-Paap rK Wald F-statistic is at the bottom of each column specification. The corresponding Stock-Yogo value for 10% (15%) maximal IV bias is 16.4 (8.96). Standard errors (in parentheses) are clustered at the firm-destination level. Key: ** significant at 1%; * 5%; * 10%.

Table G.2. Main Specification: Accounting for the Domestic Deposit Rate

Dependent Variable:	Trade Credit Share				Trade Credit Maturity			
Specification:	OLS		IV		OLS		IV	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\ln(\text{Markup})$	0.851 (0.885)	—	4.350 (5.066)	—	4.478* (2.537)	—	3.998 (11.282)	—
$\ln(\text{Markup}) \times (r_b^* - r_d)$	0.292** (0.144)	.301** (0.148)	1.260** (0.588)	1.290** (0.590)	0.188 (0.299)	0.480 (0.403)	2.732* (1.537)	3.412** (1.675)
First Stage F-Statistic	—	—	173.3	49.5	—	—	145.9	44.7
Firm FE	Yes	No	Yes	No	Yes	No	Yes	No
Product FE	Yes	No	Yes	No	Yes	No	Yes	No
Destination-Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm-product-year FE	No	Yes	No	Yes	No	Yes	No	Yes
Observations	88,546	88,546	88,546	88,546	77,328	77,328	77,328	77,328

Notes: The table replicates table 3 when interacting the markup with the difference between the foreign borrowing rate and the domestic deposit rate. All regressions are run at the firm-product-destination level (with products defined at the HS8-level). Trade credit share corresponds to the ratio of the FOB value of trade credit transactions to the FOB value of all export transactions over a year. Markups are computed at the firm-product level (products are defined at the 5-digit CPC level). Columns 1-3 report OLS, while columns 4-6 report IV results using TFPQ (and its interaction with the interest rate spread) as an instrument for markups (and its interaction with the interest rate spread). All IV regressions report the (cluster-robust) Kleibergen-Paap rK Wald F-statistic; the corresponding Stock-Yogo value for 10% maximal IV bias is 16.4. All regressions control for the logarithm of firm employment. Standard errors (in parentheses) are clustered at the firm-destination level. Key: *** significant at 1%; ** 5%; * 10%.